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NEP 2020 FOLLOWS NCF-SE

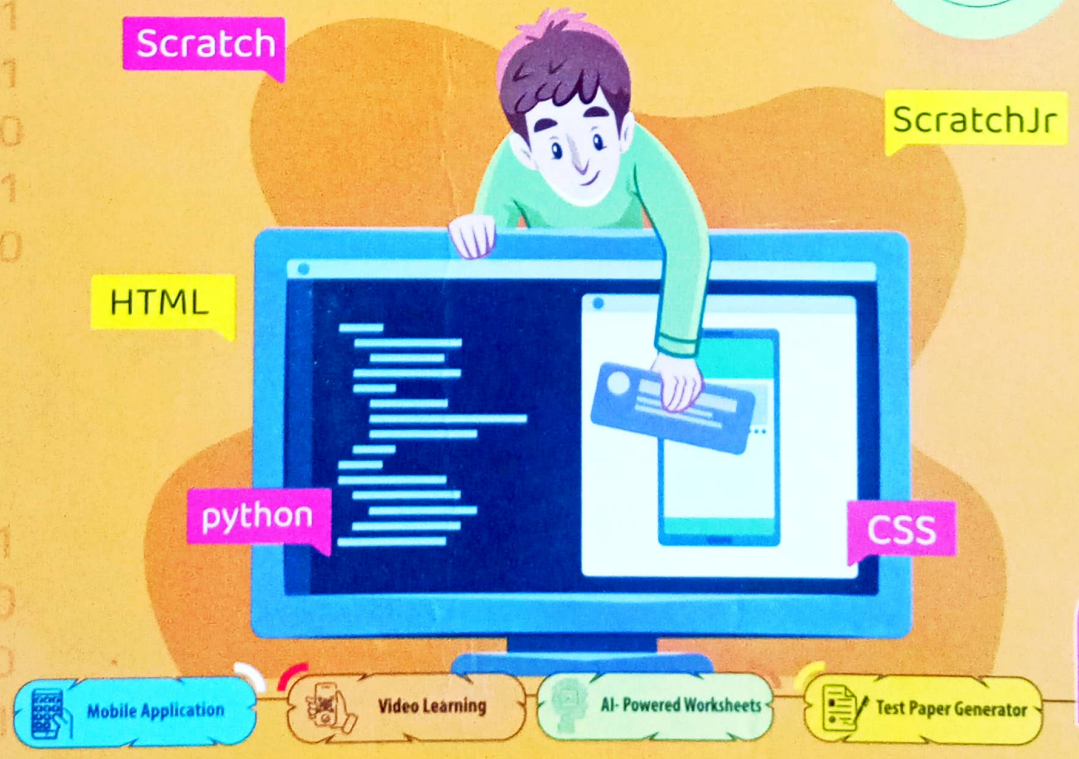
A Textbook of
Computer Science with Lab Activities

Mastering CODING

AI AND ROBOTICS

Windows 10
& MS Office 2016

PREPARATORY STAGE 3



Mobile Application | Video Learning | AI- Powered Worksheets | Test Paper Generator

Scan Code for Hybrid Learning
TOKEN
58313

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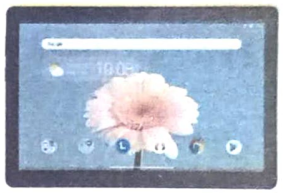
CHAPTER 1

The Computer System

Let's Recall

Write the names of the computer system shown below :



In this Chapter

- How does a Computer Work?
- Types of Computer

A computer is an electronic device that helps us perform various tasks quickly and easily. It can store and process information, allowing us to write documents, play games, and browse the internet. Computers are made up of different parts like a monitor, CPU, keyboard and mouse that work together to help us complete our work.

They are an important part of our daily lives, making many tasks simpler and more efficient.

HOW DOES A COMPUTER WORK?

A computer works using a process called the IPO cycle, which stands for Input, Process, and Output.

Input

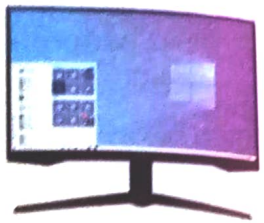
This is when the computer receives data or instructions from input devices like the keyboard or mouse. For example, when you type on the keyboard, you are giving input to the computer.

Process

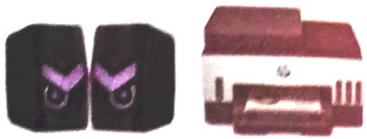
After receiving the input, the computer's CPU (Central Processing Unit) processes the data into meaningful information. The CPU is called the brain of the computer. It performs calculations, follows instructions, and makes decisions based on the input. This conversion of data into meaningful information is called processing.

Output

Once the computer processes the information, the result we get back is called output. This output is displayed on devices like the monitor or printer. The output we see on the monitor is called a soft copy. The printout of the soft copy on paper is called a hard copy.



This cycle of input, processing, and output happens very quickly, allowing the computer to perform tasks efficiently.



Activity Corner

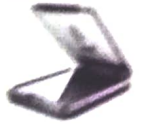
Look at the given pictures and fill the missing letters.

1. [L][ ][ ][T][ ][P]

2. [ ][E][S][ ][T][ ][P]

3. [P][ ][ ][N][T][E][ ]

4. [ ][C][ ][N][ ][ ][R]



LET'S GET BETTER

Ankit's father has asked him to take out 40 copies of the birthday invitation letter. Which device should he use to get the hard copy?

TYPES OF COMPUTERS

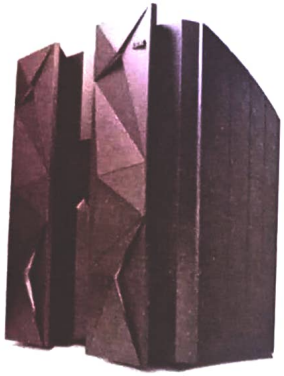
1. Microcomputers

- These are the most common type of computers used by individuals and small businesses.
- Examples of microcomputers include desktop computers, laptops, tablets, and smartphones.
- They are used for everyday tasks like browsing the internet, playing games, and completing office work.

2. Minicomputers

- Minicomputers are more powerful than microcomputers but smaller than mainframe computers.
- They are used by medium-sized businesses for tasks like managing databases and supporting multiple users.

- They are less common today but were once popular in organisations needing more computing power than a microcomputer.



3. Mainframe Computers

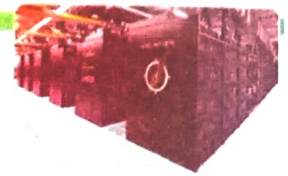
- Mainframe computers are very large and powerful machines used by big organisations.
- They can process and store vast amounts of data and support many users at the same time.
- They are used by banks, government agencies, and large companies for critical applications like transaction processing.

4. Supercomputers

- Supercomputers are the most powerful type of computers, capable of processing enormous amounts of data at incredible speeds. They are used for complex tasks like weather forecasting, scientific simulations, and research in fields like physics and astronomy.
- Supercomputers are extremely fast and can perform billions of calculations per second.
- The PARAM 8000 was India's first supercomputer, developed in 1991.
- The current supercomputer in India is the AI Supercomputer AIRAWAT.

INFO Byte

As of December 2023, 'Frontier', located in Tennesse, United States, is the world's fastest supercomputer.



I. Tick (✓) the correct option :

1. Which of the following is known as the brain of a computer?
   (a) CPU (b) Printer
   (c) Keyboard (d) Monitor
2. Which was India's first supercomputer?
   (a) Pratyush (b) PARAM
   (c) Server (d) AIRAWAT
3. A computer works on the ..... cycle.
   (a) PIO (b) IPO (c) IOP (d) POI
4. The data or instruction we give to the computer is called ..... .
   (a) input (b) output
   (c) storage (d) None of these

II. Write 'T' for true and 'F' for false :

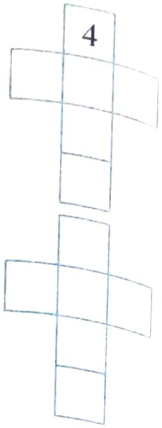
1. Smartphone is a type of minicomputer.
2. CPU does all the processing.
3. The output which is displayed on the monitor is called hard copy.
4. Input is the result we get after processing.

III. Answer the following questions :

1. What is a CPU?
2. Write the names of any two input devices.
3. What do you mean by output? Write the names of any two output devices.
4. What are the Supercomputers?

Smart Thinking

1. Write the numbers 2, 3, 5, 6 and 7 in correct place so that each line of the cross adds up to 15.



2 3 5 6 7

Total must be 15.

2. Write the numbers 2, 3, 4, 5, 6 and 7 in the squares so that each line of the cross adds up to 17.

2 3 4 5 6 7

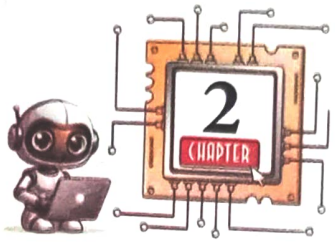
Total must be 17.

Lab Activity

Look at the computer system in your lab. Find out which parts are used for input, process and output.

FOR THE TEACHER

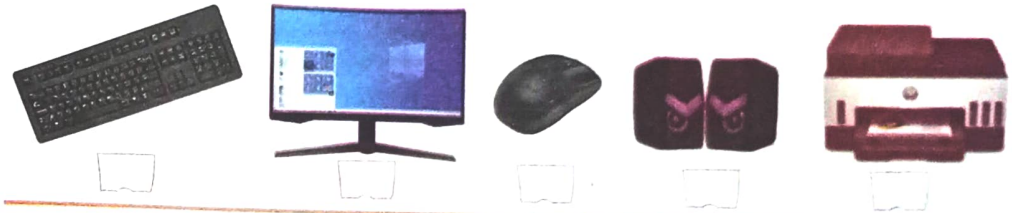
- Talk about CDAC Pune which makes Indian supercomputers like Param.
- Show children different types of Input and Output devices.



Hardware and Software

Let's Recall

Write 'I' for Input and 'O' for Output devices :



In this Chapter

- Hardware
- Types of Hardware
- Software
- Types of Software

Computers are amazing machines that help us do many things. They consist of two components : hardware and software.

HARDWARE

Hardware refers to all the physical parts of a computer that you can see and touch. These are the components that make up the computer and allow it to function. Some examples of hardware components are monitor, keyboard, mouse, CPU, printer and speakers.

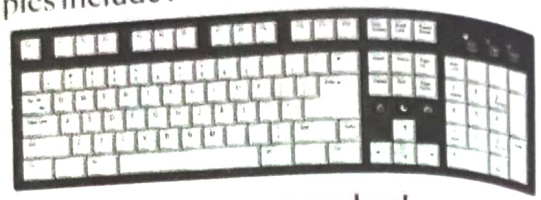
TYPES OF HARDWARE

Hardware

Input devices | Processing devices | Output devices | Storage devices

Input Devices

Input devices are hardware components that allow you to provide data and commands to the computer. Examples include :



Keyboard : Used to type letters, numbers and symbols.

Mouse : Used to move the cursor and select items on the screen.

Microphone : Captures sound and converts it into digital data.

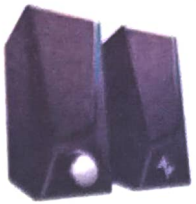
Scanner : Scans documents and images to create digital copies.

Output Devices

Output devices are hardware components that receive data from the computer and present it to you in a readable or usable form. Examples include :

Monitor : Displays visual output from the computer.

Printer : Produces paper copies of digital documents and images.



Speakers : Output sound so you can hear audio from the computer.

INFO BYTE

A projector is an output device that shows what's on the computer screen onto a bigger screen, making it easier for more people to see.

Storage Devices

Storage devices are hardware components used to save and retrieve data and files. Examples include :

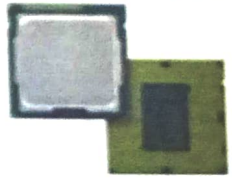
Hard Drive : A device that stores large amounts of data permanently.

USB Flash Drive : A portable device used to transfer and store files.

CD/DVD : Discs used to store and play data, music or videos.

Processing Devices

Processing devices are hardware components that handle and process data. Examples include :



CPU (Central Processing Unit) : The main processor that carries out instructions and performs calculations for the computer.

GPU (Graphics Processing Unit) : A special processor that helps to create and display images and videos.

SOFTWARE

Software refers to the set of instructions and programs that tell the computer hardware what to do. It enables the computer to perform specific tasks and functions, making it possible to interactwith and use the hardware.

TYPES OF SOFTWARE

There are three types of software. They are :

- System Software
- Application Software
- Utility Software

System Software

System Software is a type of software that helps the computer's hardware work properly. It acts as a bridge between the hardware and other software. The most common system software is the Operating System.

Operating systems are the software that manages the computer's hardware and provides a platform for other software to run. For example : MacOS, Linux, Windows etc.

System software ensures that everything on your computer works smoothly and efficiently.

Application Software

Application Software is the type of software that is used to perform specific tasks on your computer. It includes programs like :

Word Processors : They are used for writing and editing documents (e.g., Microsoft Word).

Games : They are used for entertainment and fun (e.g., Candy Crush, minecraft).

Educational Programs : They are used for learning new things (e.g., language learning apps).

Application software helps you do things like write letters, play games, or learn new skills.

Ankit wants to throw a birthday party for his friends. He wants to invite them by writing an invitation letter. Which application software will he use to write the letter?

Utility Software

Utility software helps keep the computer running smoothly by performing specific maintenance tasks. One of the common examples of utility software is antivirus software.

Difference between Hardware and Software

Hardware	Software
Physical components you can touch (e.g., keyboard, monitor).	Programs and instructions you can not touch (e.g., Windows, games).
Tangible and has a physical presence.	Intangible and exists as code.
Manufactured in factories.	Created by writing code.
Can wear out or break over time.	Doesn't physically wear out but can become outdated.
Performs tasks based on software instructions.	Provides instructions for hardware to perform tasks.

Hardware and software work together to make a computer function. Hardware needs software to operate, and software needs hardware to run. For example, without software, the hardware would be unable to perform any tasks, and without hardware, the software would have no platform to run on.

Exercise

I. Tick (✓) the correct option :

1. Which of the following is an input device?
   (a) Keyboard ☐ (b) Printer
   (c) CPU ☐ (d) Speaker
2. Which of these displays the output from the computer screen onto a larger screen?
   (a) Projector ☐ (b) Speakers
   (c) Printer ☐ (d) Scanner
3. ..... is usually fixed inside a digital camera.
   (a) Pen drive ☐ (b) Hard disk
   (c) Memory card ☐ (d) None of these
4. Which of the following is used to draw and colour the pictures?
   (a) Word ☐ (b) Paint ☐ (c) Excel ☐ (d) Notepad
5. Which of these is an application software?
   (a) Windows 7 ☐ (b) Windows 10
   (c) Word ☐ (d) None of these

II. Fill in the blanks with the help of the given box :

1. The ..... coordinates between all the parts of a computer.
2. ..... is known as the brain of computer.
3. ..... shows us the result after processing.
4. A computer accepts data and instruction from you through ..... .

CPU
CU
input devices
monitor

III. Answer the following questions.

1. What is hardware?
2. What is an operating system? Name any one operating system used in your school's computer lab.
3. What are processing devices?
4. What is system software? Given an example.
5. What is application software. Give an example.

Smart Thinking

Circle the odd one out :

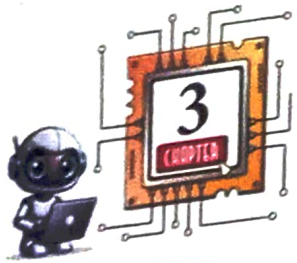
1.	Paint	Word	Windows 10	Adobe Photoshop
2.	Mouse	Scanner	Keyboard	Windows XP
3.	CD	Pen drive	Hard disk	Keyboard

Lab Activity

1. Open Notepad and type the names of various input, output and storage devices.

Input Devices	Output Devices	Storage Devices
(a) Keyboard	(a) Monitor	(a) CD
(b) Mouse	(b) Printer	(b) DVD
(c) Webcam	(c) Projector	(c) Pen drive

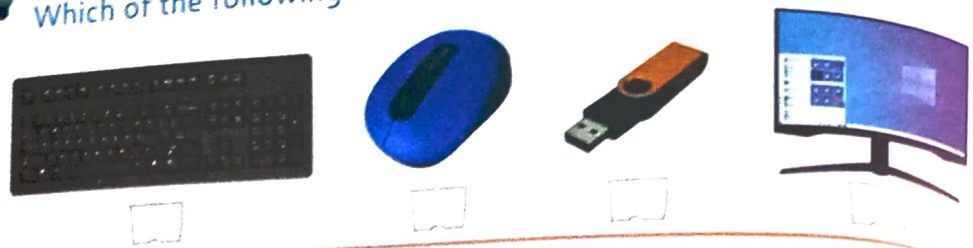
2. Open the calculator software on your computer by clicking on :
   Start → All Programs → Accessories → Calculator
   Solve the following :
   (a) 25 + 35 ..... (b) 56 - 4 .....
   (c) 13 × 6 ..... (d) 482 ÷ 167 .....



File Management

LET'S RECALL

Which of the following is a storage device?



IN THIS CHAPTER

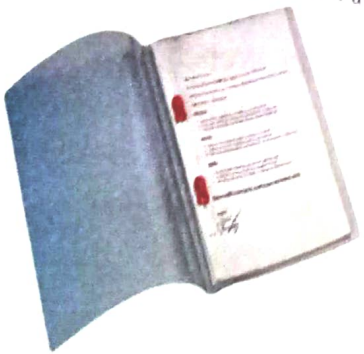
- What is a File?
- Need for Saving a File
- File in a Computer System
- Saving a File

WHAT IS A FILE?

We all write poems, stories or draw pictures on different pages of a notebook. Each page of your notebook is like a file because it contains a piece of information you created.

In the picture shown, important papers are clipped together. These papers are also like files because they hold different pieces of information.

So, we can say that a file is a page or paper that contains any piece of information.



FILE IN A COMPUTER SYSTEM

Just like the pages in your notebook, a file on a computer is a piece of information you create. You use software on the computer to make these files. A file can be a picture, a song, a document, a video, or anything else you create and save.

Here are some examples of different types of files you can create on a computer:

Poem : This could be saved as a Word file.

Result : This might be saved as an Excel file.

Birthday : This could be a PowerPoint file with pictures and text.

School Prayer : This might be an audio file.

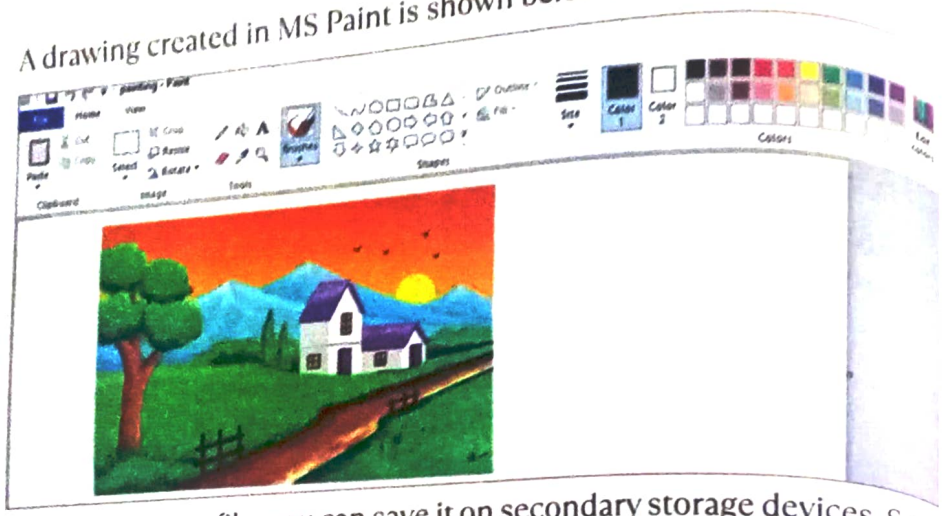
Dance : This could be a video file.

Files on a computer help you keep all your work organised and easy to find, just like the pages in your notebook. Thus, a file is information you create on a computer using software. For example, you can draw a picture in MS Paint, and that drawing is saved as a file. Here's an Example :

INFO Byte

If you delete a file by mistake, sometimes you can find it in the Recycle Bin on your computer and bring it back.

A drawing created in MS Paint is shown below :



Once you create a file, you can save it on secondary storage devices. Some secondary storage devices are :

- Pen drive
- Memory cards
- Compact Disc (CD)

These storage devices help you keep your files safe and easy to access.

NEED FOR SAVING A FILE

You need to save a file for several reasons :

- To use it in the future when you need it.
- To make changes to the file.
- To complete the file if it isn't finished yet.

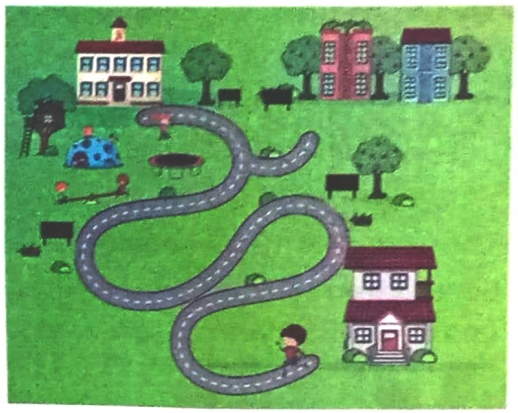
In this chapter, you will learn about file management. This means understanding how to save and organise files on a computer.

File Management is the process of saving and arranging files on a computer. It helps you keep your files organised so you can easily work with them.

Path

A path is a route or set of directions that shows you how to get from one place to another. It helps you find the way to reach your destination.

Imagine you want to go from your house to your friend's house. The path would be the streets and turns you take to get there.



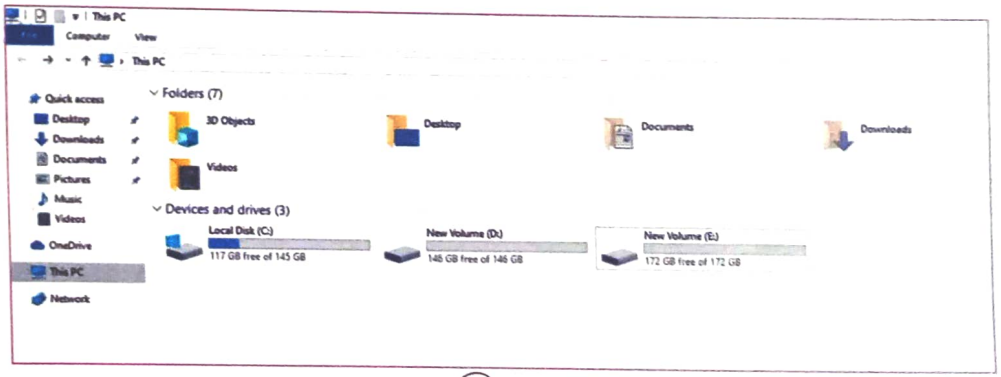
Here's a simple example of a path : House → Main Street → Park → Friend's House

In this example :

1. You start at your House.
2. You walk down Main Street.
3. You pass by the Park.
4. Finally, you arrive at your Friend's House.

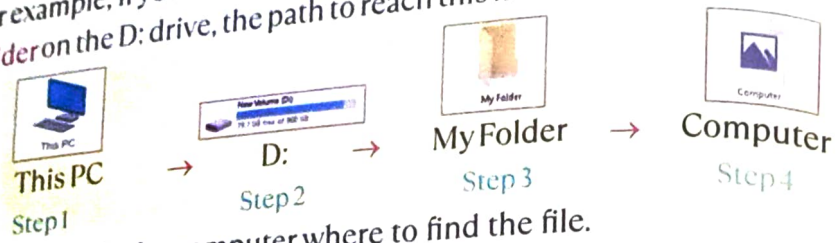
So, the path is the way you follow to reach your friend's house. It helps you know exactly where to go.

Similarly, in a computer, a path shows where to save a file in a specific location. Here's how it works : The hard disk of a computer is divided into different drives, like C:, D:, and E:.



Each drive further contains folders. A folder is a special place in a drive where similar files are stored.

For example, if you have a file called Computer saved in a folder named My Folder on the D: drive, the path to reach this file is :

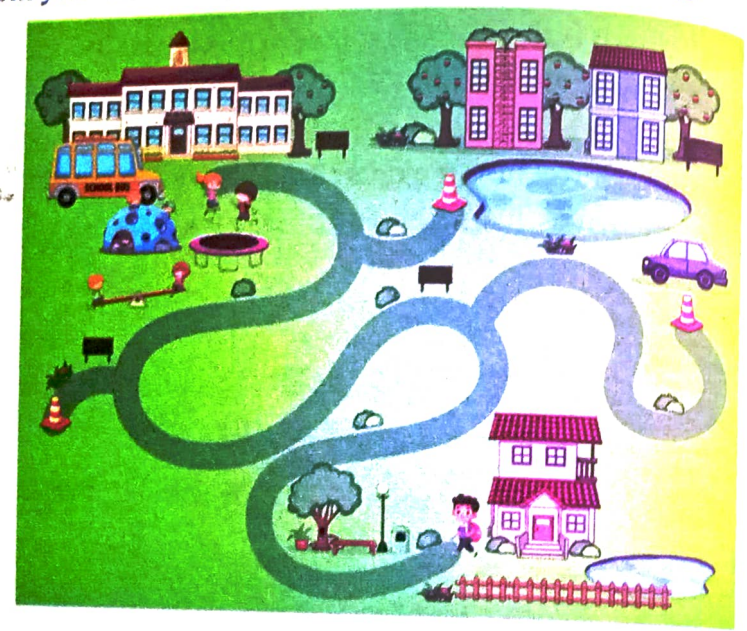


This path tells the computer where to find the file.

So, a path on a storage device is the route you follow to find a specific folder on a drive where you can save or access a file.

Activity Corner

Trace a path from your house to your school with the help of the given pictures.



SAVING A FILE

You can save a file in these places :

1. On the hard disk drive
2. In a folder on the hard disk drive
3. On a pen drive

Let's Get Better

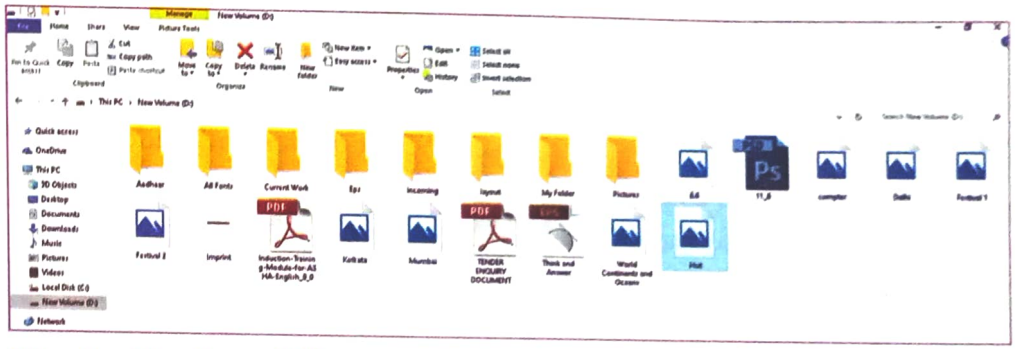
Why is it important to save your work on a computer? What might happen if you don't save it before turning off the computer?

Saving a File to the Hard Disk Drive

To save a file on the hard disk drive, follow these steps :

1. Create a file using any software, like MS Paint.
2. Click on the File tab, and then select Save As from the menu.
3. A dialogue box will open.
4. Choose the drive where you want to save the file (e.g., D: drive).
5. Enter a name for the file (e.g., Hut) in the File Name box.
6. Click Save to store the file in the D: drive.

The file will be saved in the D: drive.

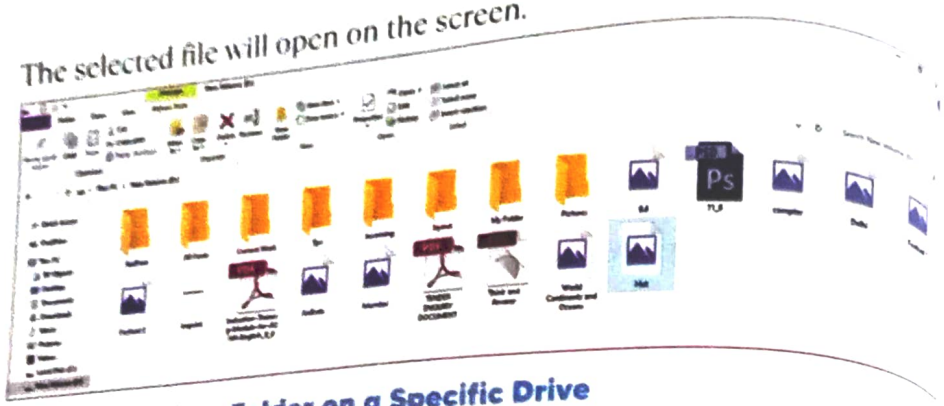


Viewing the Saved File

To view the saved file, follow these steps:

1. Click on the This PC icon on your desktop to see a list of drives.
2. Double-click on the D: drive to open it and see a list of files.
3. Find and click on the file (e.g., Hut) to open it on the screen.

The selected file will open on the screen.

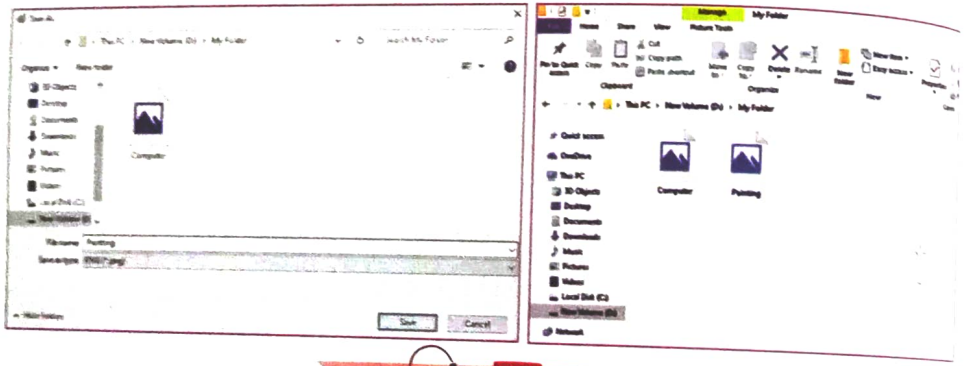


Saving a File in a Folder on a Specific Drive

A storage device or drive contains folders where you can store files. By default, files are saved in the **Documents** folder.

For example, if you create a file in MS Paint of a drawing with name Painting and you want to save it in a folder called My Folder on the D: drive, follow these steps:

1. Click on the File tab. A menu will appear.
2. Select **Save As** from the menu, then click **Browse**. This opens the Save As dialog box.
3. Double-click on the **D: drive** to see the list of files and folders.
4. Double-click on the folder named My Folder.
5. Enter a name for the file in the **File Name** box. Click Save.



Your file will now be saved in the My Folder folder on the D: drive. To find it, click on the D: drive and then open My Folder. You will see the file named painting inside it.

Saving a File to a Pen Drive

A pen drive is a small, portable storage device that you can easily carry with you. You can save files to a pen drive and take them to different places.

Here's how to save a file to a pen drive:

1. Click on the File tab in your software. A menu will appear.
2. Choose Save As from the menu, then click **Browse**. This opens the Save As dialog box.
3. Double-click on the pen drive (usually labeled as **Removable (F:)**) on the left side of the dialog box.
4. Type a name for the file (e.g., **painting**) in the **File Name** box.
5. Click Save.

Your file will be saved on the pen drive. Make sure the pen drive is plugged into a USB port on your computer while you're saving the file.

INFO Byte

Every file has a special ending, called a **file extension** (like .jpg, .mp3, .txt), that tells the computer what type of file it is.

Exercise

I. Tick (✓) the correct option:

1. Which type of file does this following clip illustrate?

   (a) Audio file ☐ (b) Video file
   (c) Dance file ☐ (d) A film

2. A file is stored in :

   (a) Primary storage ☐ (b) Secondary storage
   (c) Data storage ☐ (d) Input storage

3. Which of the following is not a storage device?

   (a) CD ☐ (b) Pen drive
   (c) Hard disk ☐ (d) Laser printer

4. To save a file, enter file name in 'Save as' dialog box and click :

   (a) Save button ☐ (b) Store button
   (c) Create button ☐ (d) Restore button

5. What is incorrect for storing a file?

   (a) To use it in future
   (b) To make changes in the file
   (c) To complete the file if it is incomplete
   (d) To delete it from the hard disk

6. To make a change in the existing file, you must ..... the file.

   (a) open ☐ (b) close
   (c) save ☐ (d) restore

II. Write 'T' for true and 'F' for false :

1. File is a storage device. ☐
2. A file can store songs and pictures. ☐
3. The files are stored only on a hard disk. ☐
4. The name of the default folder is Documents. ☐
5. Removable disk means removing hard disk from computer. ☐

III. Fill in the blanks with the help of the given box :

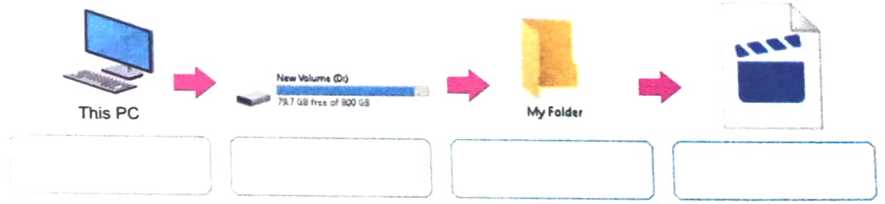
1. ..... is a route or a way to reach to a place.
2. ..... is a portable storage device.
3. Pen drive is always inserted to the ..... port.
4. A ..... stores various files.
5. Save As option is available under ..... tab.

Folder
USB
Path
File
Pen Drive

IV. Label the following to store an audio file (Song 1) and a video file (Video 1) in a specified path :

1. to a removable disk

2. to a folder (say, My Folder) which is in D: drive

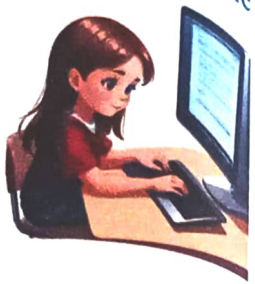


Smart Thinking

You have prepared a document in computer. Now, perform the following tasks.

(i) Give a suitable file name to your document.

(ii) Trace a path to store your file in a pen drive.



Lab Activity

- Open a drawing software like MS Paint.
- Create a simple drawing (e.g., a house, tree, or smiley face).
- Save the file with the name 'My Drawing' in the 'Documents' folder.
- Now, create another drawing and save it in a new folder on the D: drive called 'Art Projects'.
- Finally, insert a pen drive into your computer and save a copy of one of the drawings to the pen drive.

FOR THE TEACHER

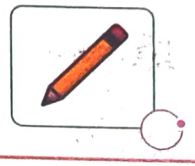
- Allow students to practice saving and organizing files by using software like MS Paint or a word processor. This will help them understand the importance of file management firsthand.
- Demonstrate the process of saving a file and navigating through different storage devices to reinforce learning. Let students explore saving files on different drives (e.g., hard disk and pen drive).

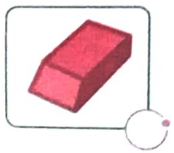
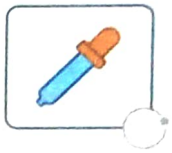
CHAPTER 4

Fun with Paint

Let's Recall

Which of the following is the icon of eraser tool in MS Paint?



In this Chapter

- MS Paint

In class 1, you learned the basics of MS Paint. This program lets you draw and paint different kinds of pictures, helping you boost your creativity and imagination. Let's quickly go over the paint tools you've learned about.

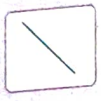
Paint Tools

The Ribbon displays all the tools available in Paint. These tools help you with drawing and colouring. You can choose these tools using your mouse. The various tools are listed in the table below:

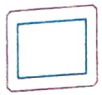
Symbol	Name of the Tool	Purpose
	Pencil	It is like a pencil which is used for free-hand drawing with the help of mouse.

Eraser — It acts like a normal eraser that you use for erasing. It is used to erase whole or part of a drawing.

Brushes — This tool provides different types of brushes with different styles such as calligraphic, crayon, marker, or water color brush.



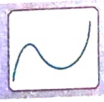
Line — You can draw straight lines within the drawing area with the help of this tool. You can also select the required thickness of the line from the options available.



Rectangle — It allows you to draw a rectangle or a square of different sizes in the drawing area.

Oval — You can draw circles or oval shapes of any size by simply dragging the mouse within the drawing area using this tool.

Color Palette — You can choose any colour of your choice from here.



Curve — It enables you to draw curved lines in the drawing area.

In this chapter, you will learn more features of Paint software that can make your drawings look more realistic.

Editing a Drawing

Editing means making changes to your drawing. Here are the different tasks you can do :

1. Select
2. Cut and Paste
3. Copy and Paste
4. Undo
5. Redo
6. Delete
7. Resize
8. Change the Drawing Area Size
9. Rotate or Flip
10. Full Screen

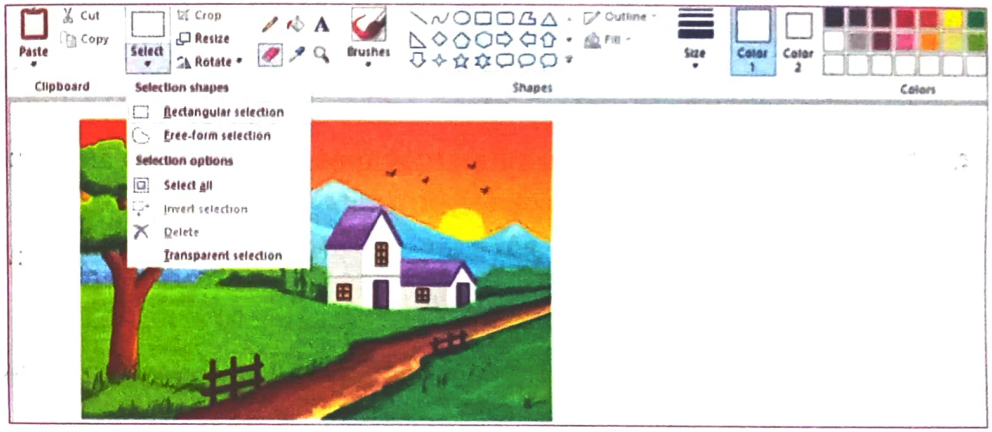
Selection

This option is used when you want to move or copy a part of your drawing. When you click on the Select option, you have two ways to make a selection:

1. Rectangular selection
2. Free-form selection

The Rectangular selection tool lets you select or move an area of the drawing in a square or rectangular shape. Follow these steps to use the Rectangular selection:

Step 1. Click on the Shapes option and choose a shape from the dropdown list.



Step 2. Draw your picture using different shapes in the drawing area. You can fill the shapes with color if you want.

Step 3. Click on the Select tool and choose Rectangular selection from the dropdown menu.

Step 4. Drag the mouse to cover the part of the drawing you want to move or copy.

Step 5. Release the mouse button. The selected part will be marked with a dotted box.

Now, you can move or copy the selected area to another part of the drawing.

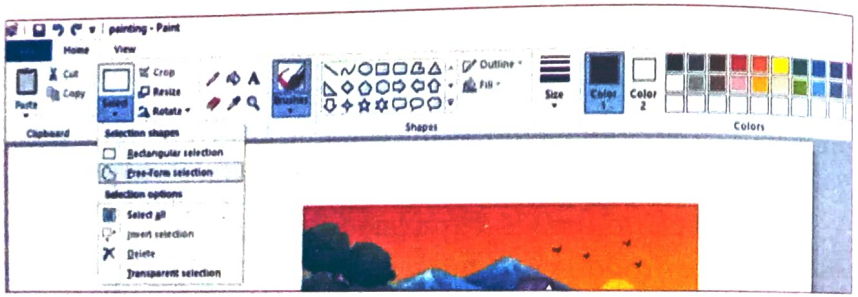
Free Form Selection

This tool helps you move or copy irregular parts of a drawing as needed. Follow these steps to use Free-form selection :

Step 1. Click on the Shapes option and choose a shape from the dropdown list.

Step 2. Draw your picture using different shapes in the drawing area. You can fill the shapes with color if you want.

Step 3. Click on the Select tool and choose Free-form selection from the dropdown menu.



Step 4. Drag the mouse to outline the area of the drawing you want to move or copy.

Step 5. Release the mouse button.

The selected part of the drawing will be marked with a dotted box. You can now move or copy this selected area to another part of the drawing.

Cut and Paste

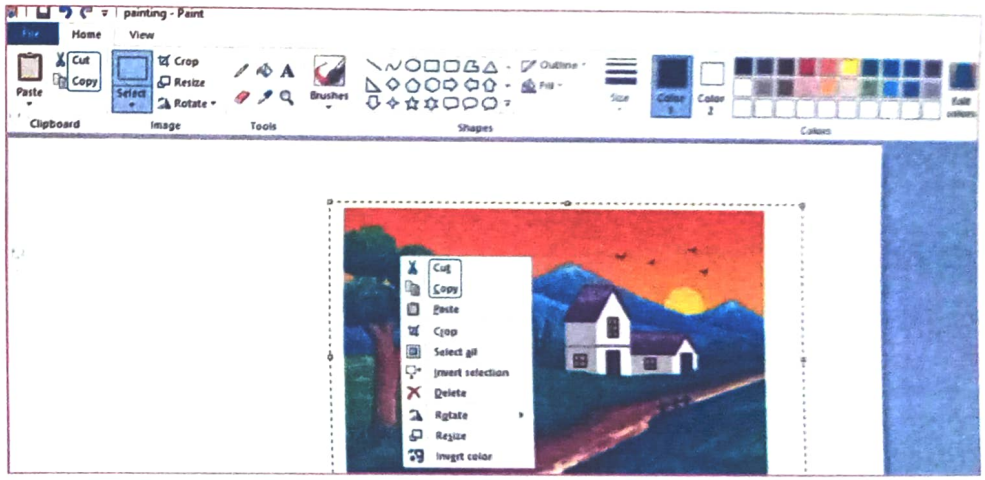
The Cut and Paste operation removes a selected part of your drawing from its original place and pastes it in a new location within the drawing area. Follow these steps to do this:

Step 1. Draw an object or open a picture file from your computer in the Paint window.

Step 2. Click on the Select option and choose Rectangular selection from the dropdown menu.

Step 3. Select the part of the drawing you want to cut. The selected area will be outlined with dotted lines.

Step 4. Click on the Cut option from the Clipboard menu or right-click on the selected area and choose Cut from the dropdown menu.



Step 5. Right-click anywhere in the Paint window and choose Paste from the dropdown menu.

The selected part of the drawing will be pasted in the top left corner of the drawing area. You can move the pasted picture by clicking and dragging it to the desired location within the drawing area.

Copy and Paste

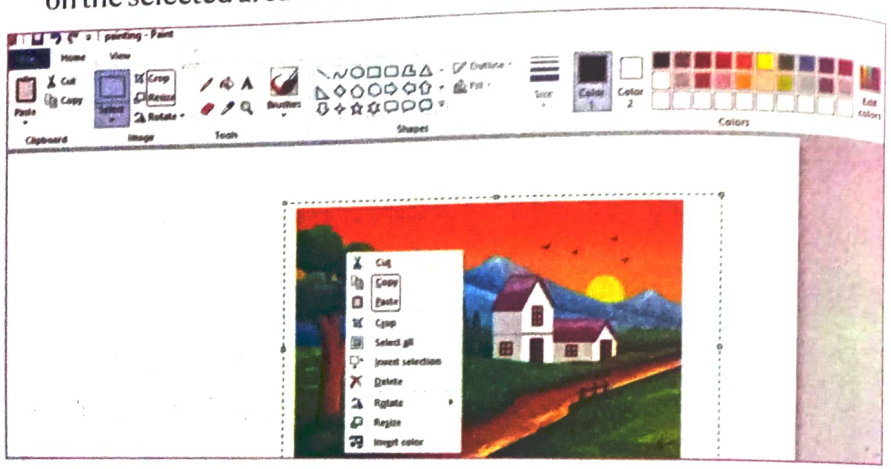
When you need to make a duplicate copy of a drawing or a selected part of your drawing, use the Copy and Paste commands. Follow these steps to do this:

Step 1. Draw an object or open a picture file from your computer in the Paint window.

Step 2. Click on the Select button and choose Rectangular selection from the dropdown menu.

Step 3. Select the part of the drawing you want to copy. The selected area will be outlined with dotted lines.

Step 4. Click on the Copy option from the Clipboard menu, or right-click on the selected area and choose Copy from the dropdown menu.



Step 5. Right-click anywhere in the Paint window and choose Paste from the dropdown menu.

The copied part of the drawing will be pasted in the top left corner of the drawing area. You can move the duplicate copy by clicking and dragging it to the desired location within the drawing area.

Undo

The Undo command lets you reverse your last action or a series of actions.

Redo

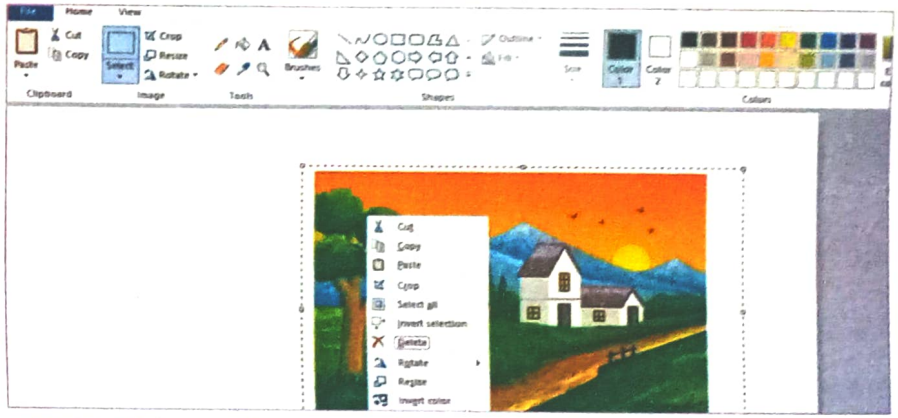
The Redo command lets you undo the effect of the Undo command, restoring your last action.

Delete

The Delete command works like the Cut command. It removes the selected part of the drawing from the drawing area. Follow these steps:

Step 1. Click on the Select option and choose Rectangular selection from the dropdown menu.

Step 2. Select the part of the drawing you want to remove. The selected area will be outlined with dotted lines.



Step 3. Right-click on the selected area and choose Delete from the dropdown menu.

The selected part of the drawing will be removed.

When you select any shape, it appears with dotted lines. This allows you to resize and position the shape within the drawing area. Once you click outside the object, the dotted line disappears, and the shape cannot be edited.

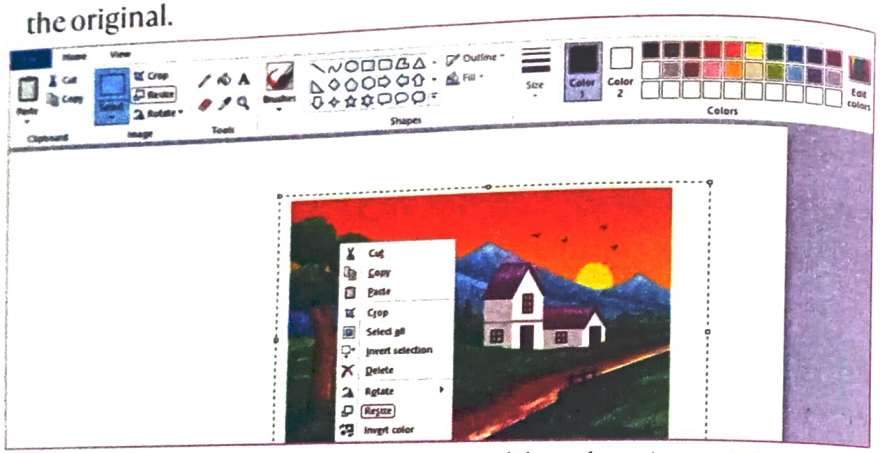
Resize

The Resize option helps you change the size of the entire image or a part of it. You can also skew an object to make it appear slanted. Follow these steps to resize a drawing:

Step 1. Open your drawing in the drawing area.

Step 2. On the Home tab, in the Image group, click on the Resize option.

Step 3. A Resize and Skew dialog box will appear. Choose Percentage to scale your image by a certain percent of its current size, or Pixels to set it to a specific width or height. Check the Maintain aspect ratio box to keep the resized picture in the same proportion as the original.



Step 4. In the Resize area, enter a new width value (e.g., 40) in the Horizontal box or a new height in the Vertical box.

Step 5. Click the OK button.

The object will be resized in the drawing area.

If the Maintain aspect ratio box is checked, you only need to enter either the width (Horizontal) or height (Vertical) value. The other box will update automatically.

INFO BYTE

If Maintain Aspect Ratio check box is selected, you only need to enter the horizontal value (width) or vertical value (height). The other box in the resize area is updated automatically.

Resizing the Drawing Area

You can resize the drawing area using one of these methods:

Method 1

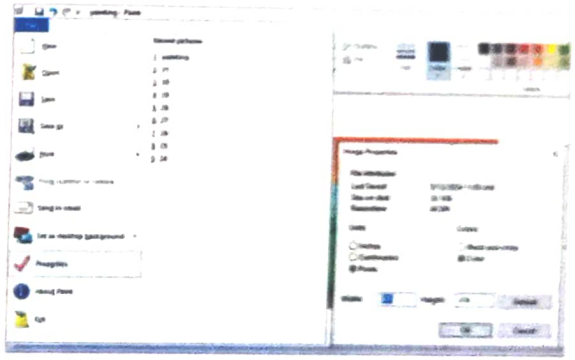
- Point the mouse to the small square at the bottom-right corner of the drawing area.
- Click and drag the square inward or outward to decrease or increase the size of the drawing area.

Method 2

Step 1. Click the File tab and select Properties.

Step 2. An Image Properties dialog box will appear. Enter the values for Width and Height in the boxes.

Step 3. Click the OK button.



The drawing area will be resized to the new dimensions.

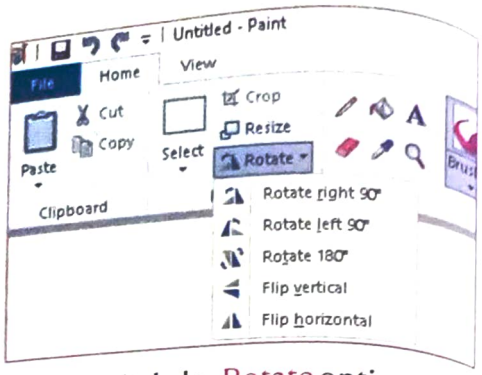
Note : When resizing the drawing area, some parts of the object or image may get cut off.

Rotate or Flip

This command lets you rotate an object in different directions or flip it. The available options are :

- Rotate right 90°

- Rotate left 90°
- Rotate 180°
- Flip vertical
- Flip horizontal



To use these options follow these steps :

Step 1. Select the object or picture.

Step 2. On the Home tab, in the Image group, click the Rotate option.

Step 3. Choose the direction you want from the dropdown menu. The object will rotate accordingly.

You can also choose Flip vertical or Flip horizontal to flip the object in the desired direction.

INFO Byte

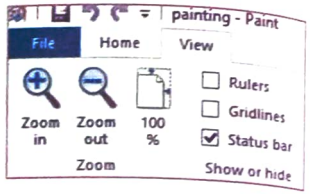
All the movements of the object are rotated or flipped with respect to the previous position.

Full Screen

You can view your picture or drawing in full screen mode. Here's how :

Step 1. Click on the View tab.

Step 2. Click the Full screen option. Your drawing will now be displayed in full screen.



To exit full screen and return to the Paint window, simply click on the drawing or picture.

Saving a File in Different Formats

After finishing your work, you can save your drawing in various file formats like *.png, *.jpeg, *.bmp or *.gif. MS Paint usually saves files in the .bmp format by default. To save your drawing in a different format, follow these steps:

Step 1. Click the File tab.

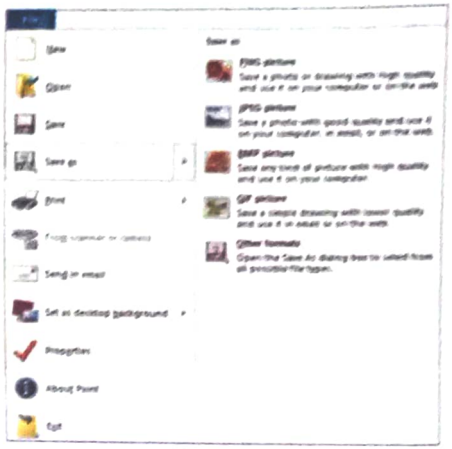
Step 2. Select Save As to see the available file formats.

Step 3. Choose the file format you want (e.g., JPEG picture).

Step 4. Enter a file name in the File Name box (e.g., House).

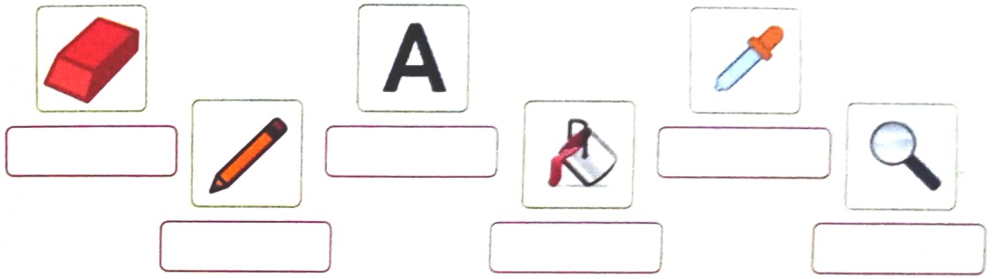
Step 5. Click the Save button.

Your drawing will be saved in the selected format for future use.



Activity Corner

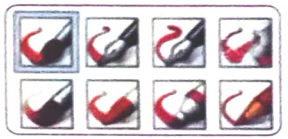
Name these tools used in Paint :



Exercise

I. Tick (✓) the correct option:

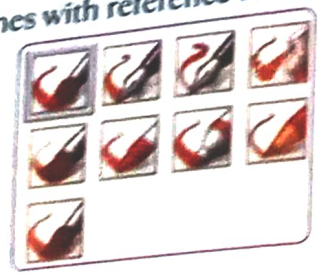
1. Which of the following tools has been depicted in the image given aside?



(a) Crayon ☐ (b) Marker ☐

(c) Water Color ☐ (d) Brushes ☐

3. Name these brushes with reference to MS Paint.



(i) .....

(ii) .....

(iii) .....

(iv) .....

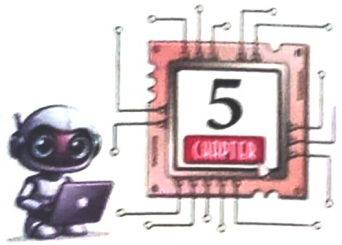
(v) .....

(vi) .....

(vii) .....

(viii) .....

(ix) .....



5 CHAPTER Introduction to MS Word

LET'S RECALL

Which of the following is the icon for MS Word?

IN THIS CHAPTER

- Features
- Starting
- Components
- Typing and Editing

Microsoft Word 2016 is a software program used for creating and editing documents. It's part of the Microsoft Office 2016 suite. With Word 2016, you can write and format text, add images, and create professional-looking documents easily. It has many tools and features to help you make your documents look great and is commonly used for tasks like writing letters, reports, and resumes.

FEATURES OF MS WORD 2016

Microsoft Word 2016 is a word processing software developed by Microsoft. It allows you to :

- Easily type letters, stories, reports, and more.
- Make changes to the text both while typing and after you're done.
- Enhance the text's appearance using various colors, styles and effects.

- Insert pictures into your document.
- Check for spelling and grammar mistakes.
- Copy and move text as needed.
- Save your work for future use.
- Print any part or the entire document.

STARTING WORD 2016

To start Word 2016, follow these steps :

Step 1. Click the Start button on the Taskbar.

Step 2. Scroll down to 'W' in the Start menu and select Word.

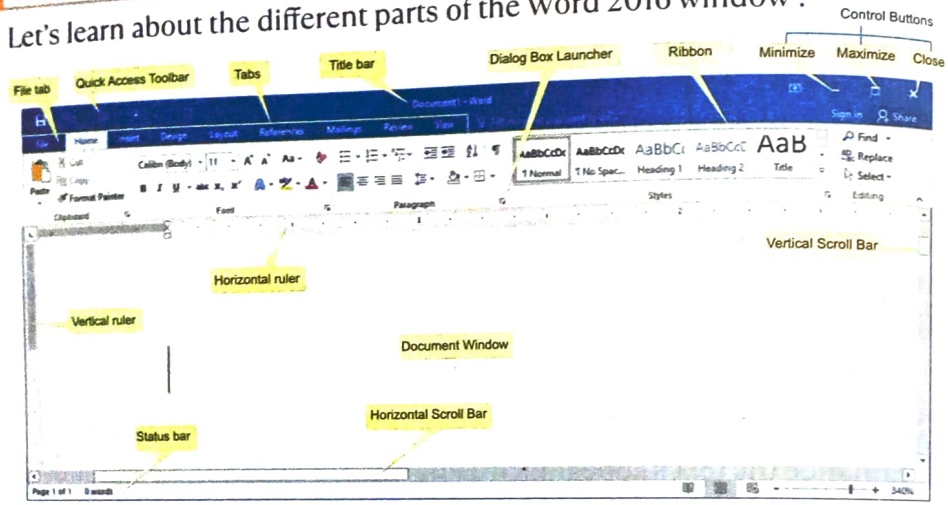
Step 3. Click on Blank document to open a new, blank document.

Now, Word 2016 will open with a new document ready for you to use.

Opening Word 2016

COMPONENTS OF THE WORD 2016 WINDOW

Let's learn about the different parts of the Word 2016 window :



Title Bar

- Displays the name of the program and the current document.
- New documents are temporarily named Document 1.
- You can rename your document when you save it.

Tabs

- Located below the title bar and Quick Access Toolbar.
- There are nine tabs : Home, Insert, Design, Layout, Reference, Mailing, Review, View and Help.
- Each tab contains related functions and commands.

Quick Access Toolbar

- Located to the left of the title bar.
- Contains frequently used buttons like Save, Undo and Redo.

Ribbon

- Located below the Quick Access Toolbar.
- Made up of tabs and groups.
- Each tab has a set of commands presented as buttons.

INFO Byte

You can pin or hide the ribbon by clicking the small arrow in the top right corner of the ribbon.

Rulers

- Horizontal ruler at the top of the document.
- Vertical ruler on the left side of the document.

Scroll Bars

- Vertical scroll bar on the right side of the screen for moving up and down.
- Horizontal scroll bar above the status bar for moving left and right.
- The horizontal scroll bar appears only if the document's width exceeds the screen width.

Text/Document Area

- The large area below the ruler where you type your content.

Cursor

- The blinking vertical line in the text area.

- Indicates where your text will appear as you type.

Status Bar

- Shows detailed information about the document, such as page number, number of pages, lines and words.

Window Control Buttons

- Located on the right side of the title bar.
- Used to minimize, maximize/restore and close the window.

Zoom Slider

- Located at the bottom right corner of the window.
- Used to quickly enlarge or reduce the size of the document view.

TYPING AND EDITING IN WORD

In Word, you create documents by typing text. When the text reaches the end of a line, the cursor automatically moves to the next line. While typing, you often use the Spacebar, Enter, Delete and Backspace keys.

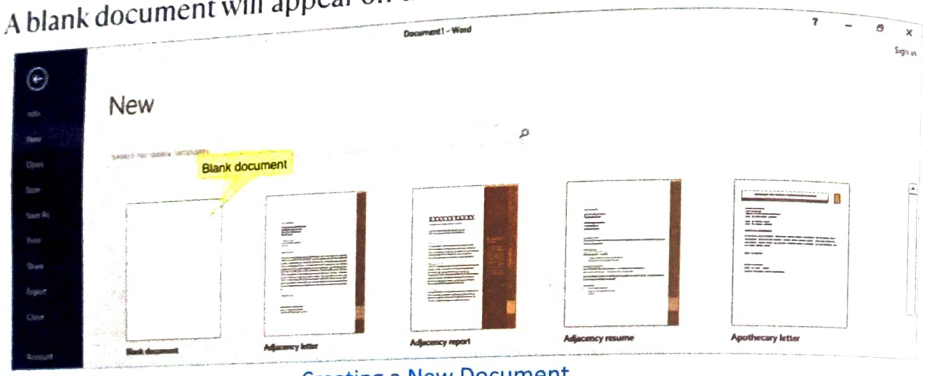
Creating a New Document

To create a new document in Word 2016, follow these steps :

Step 1. Click the File tab.
Step 2. Select New from the left pane.
Step 3. Click on Blank document.

SHORTCUT SECTION To open a new document, Ctrl + N

A blank document will appear on the screen.



Creating a New Document

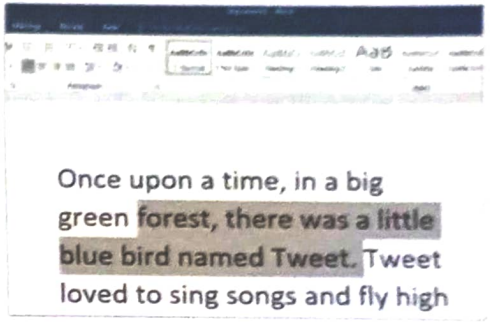
Now, let's learn about the various editing tools in Word 2016.

Typing Text

You can type text directly into the document. As you type, the cursor automatically moves to the next line when you reach the end. Press the Enter key only when you want to start a new line.

Selecting Text

To make changes to existing text, you need to select it first. Below is an example of selected text in a paragraph. Let's learn the different methods to select text.



Selecting Text Using a Keyboard

- To select a letter to the right or left of the cursor : Press Shift + Right Arrow or Shift + Left Arrow.

- To select one line up or down : Press Shift + Up Arrow or Shift + Down Arrow.

- To select a word to the right of the cursor : Press Ctrl + Shift + Right Arrow.
- To select a word to the left of the cursor : Press Ctrl + Shift + Left Arrow.

- To select the entire document : Press Ctrl + A.

Selecting Text Using a Mouse

- To select a word : Double-click on the word.
- To select a line : Click in the left margin next to the line.

- To select a sentence : Hold the Ctrl key and click on the sentence.
- To select a paragraph : Triple-click within the paragraph.

Deleting Text

If you make a mistake while typing, you can remove text using the Backspace or Delete key :

- Backspace Key : Click to the right of the cursor and press Backspace. The character to the left of the cursor will be deleted. To delete a word to the left of the cursor, press Ctrl - Backspace.

- Delete Key : Click to the left of the cursor and press Delete. The character to the right of the cursor will be deleted. To delete a word to the right of the cursor, press Ctrl - Delete.

- To remove a large amount of text, such as a sentence or paragraph, select it using any selection method and press Backspace or Delete.

Inserting Text

If you need to add more text to an already written document, you can easily do so :

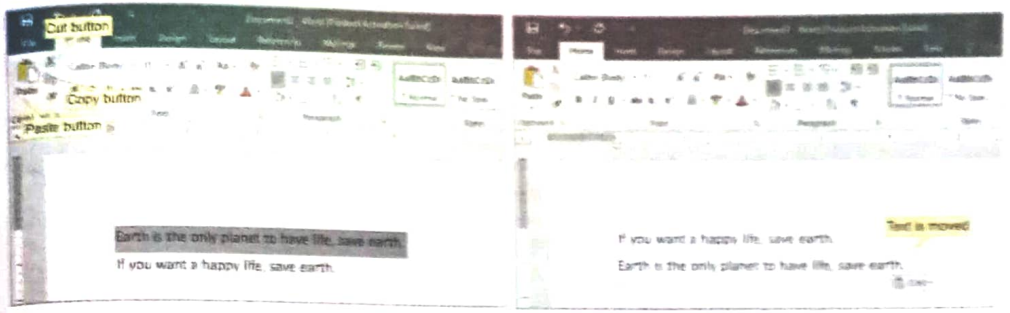
- Place the cursor at the position where you want to insert the new text and click. The insertion point (cursor) will appear at that spot.
- Type the new text, and it will be inserted at the cursor's location.

Cutting and Copying Text

Cutting text moves the selected text from its original location to a new location where you paste it. Copying text keeps the selected text in its original location and also pastes a copy at the new location.

To cut or copy text, follow these steps :

Step 1. Select the text you want to cut or copy.



Step 2. Click on the Home tab.

Step 3. Click Cut or Copy in the Clipboard group.

Step 4. Go to where you want to paste the text.

Step 5. Click Paste in the Clipboard group.

LETS GET BETTER

Rohit is making an assignment in Word. The teacher has asked him to save his document without using the mouse. Which shortcut keys will he press to save the file?

Undo and Redo

Undo cancels the last action you took. To undo, click the Undo button on the QuickAccess Toolbar.

Redo reverses the Undo action. To redo, click the Redo button on the QuickAccess Toolbar.

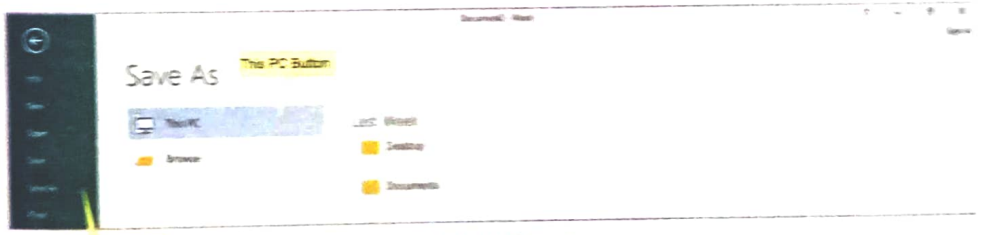
SHORTCUT To Undo : Ctrl + Z
SECTION To Redo : Ctrl + Y

Saving a Document

To save your document :

Step 1. Click the File tab.

Step 2. Select Save or Save As.

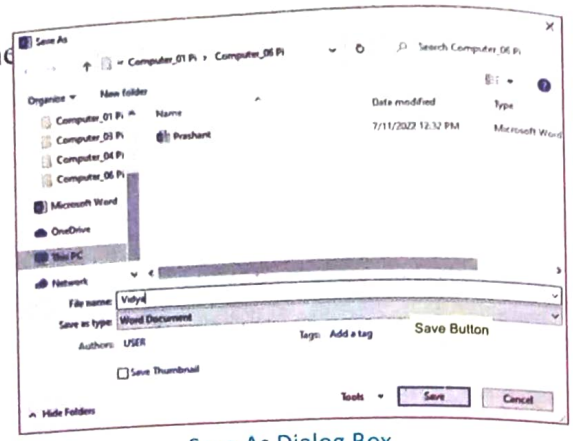


Step 3. Choose This PC and the

Step 4. In the Save As dialog box, choose where to save the file and type a name for it.

Step 5. Click the Save button.

SHORTCUT SECTION To save a document, press Ctrl + S



Save As Dialog Box

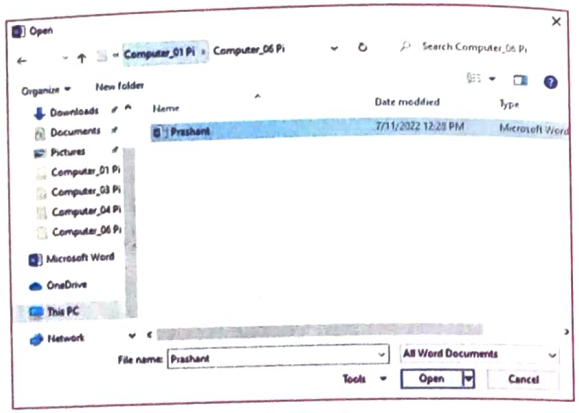
Opening a Saved Document

To open a saved document :

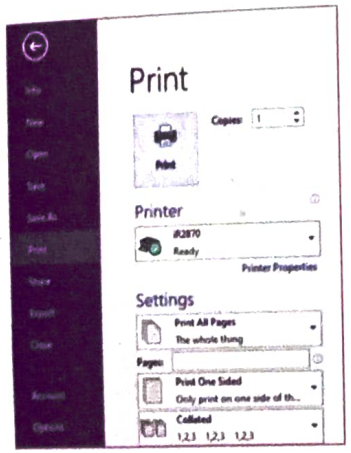
Step 1. Click the File tab.

Step 2. Select Open.

Step 3. Find the file you want to open and click the Open button.



Open Dialog Box



Printing a Document

To print your document :

Step 1. Click the File tab.

Step 2. Select Print.

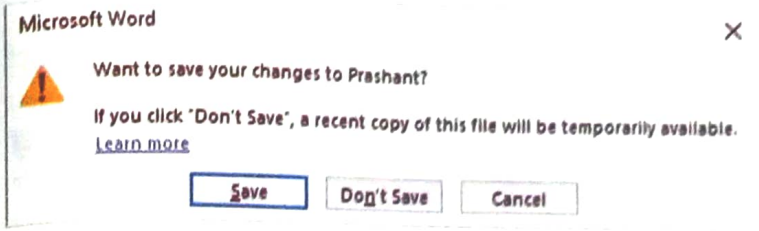
Step 3. Click the Print button to print your document.

Exiting Word

To close a document or exit Word :

Step 1. Click the File tab.

Step 2. Click Close.



Message Box

SHORTCUT SECTION To close a document, press Ctrl + F4.

Close option of Info screen

I. Tick (✓) the correct option :

1. Word 2016 is a word ..... software.
   (a) processing ☐ (b) calculating ☐
   (c) painting ☐ (d) None of these ☐
2. Which of the following bar shows the name of the program and document name?
   (a) Ribbon ☐ (b) Quick Access ☐
   (c) Title ☐ (d) Scroll ☐
3. Which of the following option is selected from the File tab to create a document?

(a) New (b) Open

(c) Save (d) None of these

4. To select a sentence, click the left mouse button while holding the ..... key.

(a) Ctrl (b) Shift (c) Alt (d) Enter

II. Fill in the blanks with the help of the given box :

zoom slider
window control buttons
file tab
editing
delete
select

1. Print option is present under the ..... .
2. ..... are used minimize, maximize or restore and close the window.
3. ..... is used to enlarge and reduce the size of window.
4. To delete large text, ... it and press ..... key.
5. Making changes in the existing text is called ..... the document.

III. Write 'T' for true and 'F' for false :

1. Word gives Document 1 as permanent file name.
2. The Ribbon contains multiple tabs.
3. Status bar is located at the bottom of Word window.
4. Cutting the text means text will disappear from its original position.
5. Undo is used to reverse the action of command.

IV. Answer the following questions :

1. Write any two features of Word 2016.
2. Write one difference between inserting text and deleting text.
3. How will you save a file in Word 2016?
4. Differentiate between copying and cutting text.

Smart Thinking

Write the shortcut command for each.

To save a document →

To open a saved document →

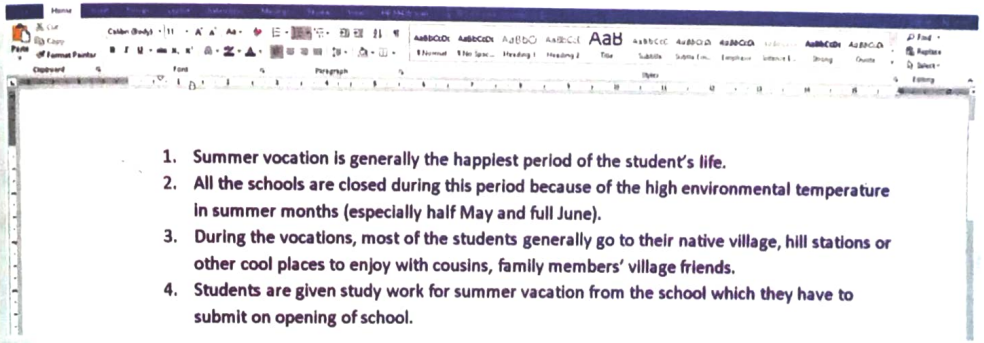
To print a document →

To close Word →

To create a new document →

Lab Activity

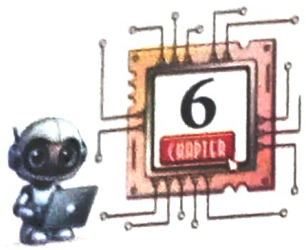
1. Write a letter to your Father telling him about your Annual Day.
   (a) Save the letter (b) Exit Word
2. Type the following sentences. Do the following :
   - Add a heading.
   - Add a sentence after Line 3. "Some of the kids join swimming or dancing classes in the vacation."
   - Move Line 4 after first line.



1. Summer vocation is generally the happiest period of the student's life.
2. All the schools are closed during this period because of the high environmental temperature in summer months (especially half May and full June).
3. During the vacations, most of the students generally go to their native village, hill stations or other cool places to enjoy with cousins, family members' village friends.
4. Students are given study work for summer vacation from the school which they have to submit on opening of school.

FOR THE TEACHER

Provide a detailed guide on using Microsoft Word 2016, including an explanation of its various components, its uses and essential shortcuts for efficient operations.



Formatting in Word

Let's Recall

Write the steps to save a word file.

.....

.....

.....

In this Chapter

- Changing Font, Font Size and Font Colour
- Bold, Italics, Underline
- Aligning the Text
- Adding bullets and numbering
- Printing a document
- Highlighting Text
- Applying Text Effects
- Changing case
- Line spacing and paragraph spacing

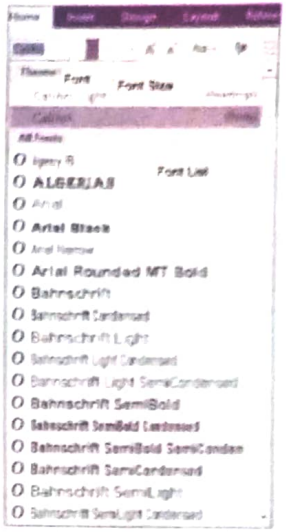
You've already learned about the features of Word 2016. Now, let's explore formatting features, which let you modify your document to give it a fresh look.

Editing refers to making changes to the existing text. Formatting involves changing the appearance and layout of the text. You can format your text using the options in the Font group.

CHANGING FONT AND FONT SIZE

Font refers to the style of the letters on the screen. Changing the font can make your document look more appealing. Word offers various fonts and sizes, with Calibri as the default font and size 11.

To change the font, follow these steps :



Step 1. Select the text you want to format.

Step 2. Click on the Home tab.

Step 3. Click the drop-down arrow next to the Font box in the Font group.

Step 4. Choose a font from the list and click to apply it.

To change the font size, follow these steps :

Step 1. Select the text you want to format.

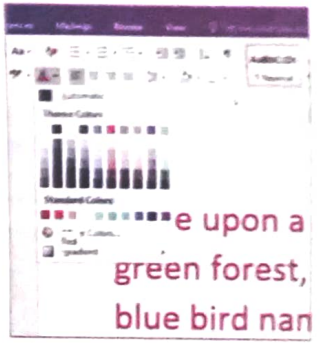
Step 2. Click on the Home tab.

Step 3. Click the drop-down arrow next to the Font size box in the Font group.

Step 4. Choose a font size from the list and click to apply it.

Changing Font Colour

You can change the colour of your text in Word. Here's how :



Step 1. Select the text you want to change the colour of.

Step 2. Click on the Home tab.

Step 3. Click the drop-down arrow next to the Font color box in the Font group. A color palette will appear.

Step 4. Choose the colour you want from the palette and click to apply it.

Highlighting Text

Highlighting text means marking important parts with a coloured background. Here's how to do it :

Step 1. Select the text you want to highlight.

Step 2. Click on the Home tab.

Step 3. Click the drop-down arrow next to the Text Highlight color button in the Font group. A color palette will appear.

Step 4. Choose a colour from the palette to apply the highlight.

Bold, Italic and Underline

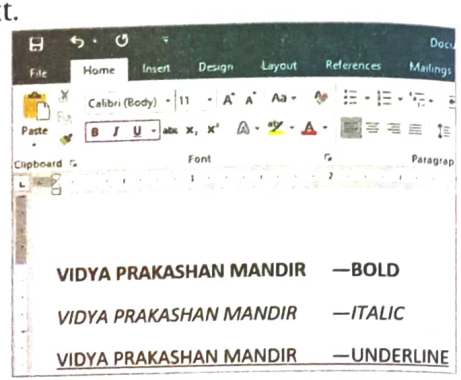
- Bold makes the text darker.
- Italic slants the text.
- Underline adds a line under the text.

To make text bold, italic or underlined :

Step 1. Select the text you want to format.

Step 2. Click on the Home tab.

Step 3. Click the **B** *I* U button in the Font group.



Applying Text Effects

Text effects add special styles to your text. Here's how to use them :

Step 1. Select the text you want to apply an effect to.

Step 2. Click on the Home tab.

Step 3. Click the Text Effects button in the Font group.

Step 4. Choose the effect you like from the options that appear.

Aligning Text

Text alignment refers to how text is positioned with respect to the margins in a document. There are four alignment options in Word:

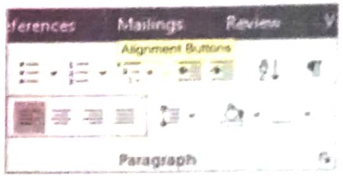
1. Align Left : Aligns the text with the left margin.
2. Align Center : Centers the text between the left and right margins.
3. Align Right : Aligns the text with the right margin.
4. Justify : Aligns the text evenly with both the left and right margins.

To change text alignment :

Step 1. Select the text you want to align.

Step 2. Click on the Home tab.

Step 3. Click one of the alignment buttons in the Paragraph group.



Changing Case

The Change Case option lets you adjust the capitalization of your text. Word offers five different case options :

1. Sentence case : Capitalizes the first letter of each sentence and leaves the rest in lowercase.
2. lowercase : Changes all letters to lowercase.
3. UPPERCASE : Changes all letters to uppercase.
4. Capitalize Each Word : Capitalizes the first letter of each word and leaves the rest in lowercase.
5. tOGGLE cASE : Switches between two case styles, like Capitalize Each Word and the opposite (e.g., cAPITALIZEeACHwORD).

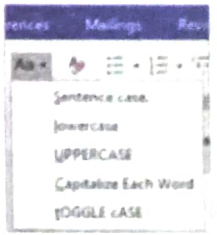
To change the case of your text :

Step 1. Select the text you want to format.

Step 2. Click on the Home tab.

Step 3. Click the Change Case option in the Font group.

Step 4. Choose the case style you want to apply.



Let's Get BETTER Kajal made a birthday card in Word in which she wrote a message using font color, font size and font type. Her mother asked her to apply some pretty cool effects to the text. Name the tab and the group Kajal should use to do this.

ADDING BULLETS AND NUMBERING

Bullets

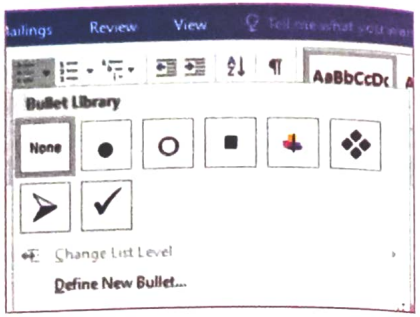
Bullets are small symbols like dots or squares used to list items. Here's how to add them:

Step 1. Click on the Home tab.

Step 2. Click the drop-down arrow next to Bullets in the Paragraph group.

Step 3. Choose the bullet style you want from the Bullet library.

Step 4. Type the first item in your list and press Enter. A new bullet will appear for the next item.



Numbering

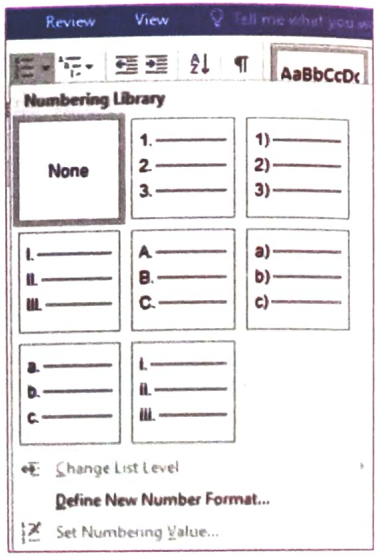
Numbering adds sequential numbers to list items. Here's how to do it:

Step 1. Click on the Home tab.

Step 2. Click the drop-down arrow next to Numbering in the Paragraph group.

Step 3. Choose the numbering style you want from the Numbering library.

Step 4. Type the first item in your list and press Enter. A new number will appear for the next item. Press Enter twice to finish the list.



LINE SPACING AND PARAGRAPH SPACING

While creating Word documents, it is important to make sure that the spacing between the lines are consistent. Word provides tools to maintain that. Let is learn more about these.

Line Spacing

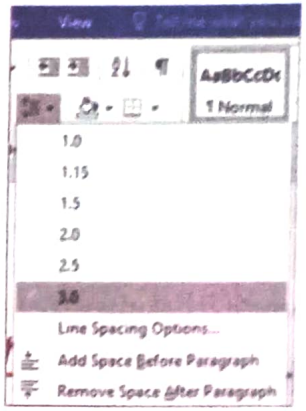
Line spacing controls the space between lines in a paragraph. To adjust line spacing:

Step 1. Select the text.

Step 2. Click on the Home tab.

Step 3. Click the drop-down arrow next to Line and Paragraph Spacing in the Paragraph group.

Step 4. Choose the spacing option you want from the menu.



The spacing between lines will change according to your selection.

INFO Byte

You can adjust the line spacing by clicking the paragraph arrow in the paragraph group under the Home tab. Then, select any line spacing style from the line spacing drop-down menu.

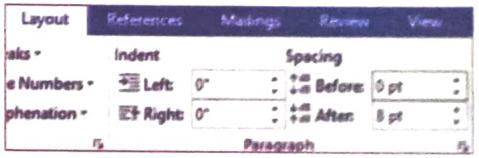
Paragraph Spacing

Paragraph spacing controls the space before and after a paragraph. To adjust paragraph spacing:

Step 1. Select the paragraph.

Step 2. Click on the Layout tab.

Step 3. In the Paragraph group, find the Spacing section.



Step 4. Adjust the values in the Before and After boxes to change the spacing around the paragraph.

INFO Byte

You can also adjust the paragraph spacing by clicking the paragraph arrow in the paragraph group under the home tab. In the Spacing section, enter values for spacing before and after the paragraph.

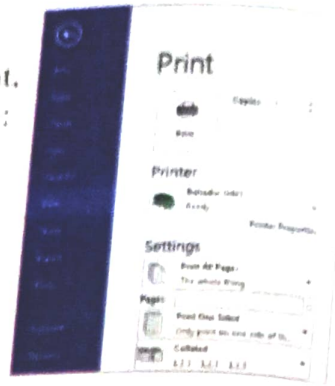
Printing a Document

Once you've finished formatting your document, you can print it to get a physical copy. Here's how:

Step 1. Click on the File tab.

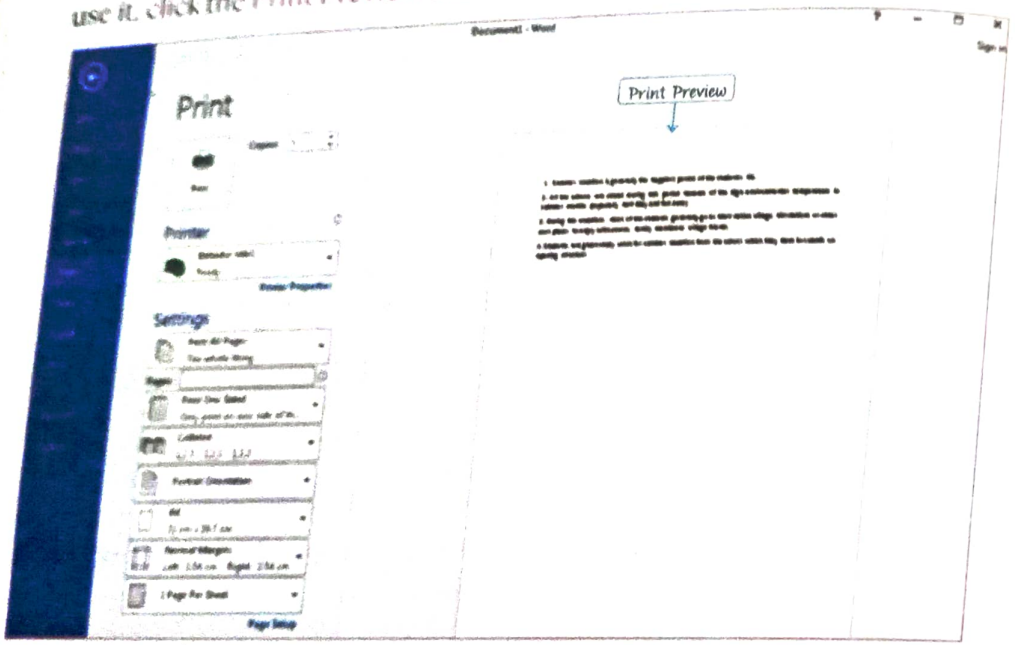
Step 2. Select Print from the menu.

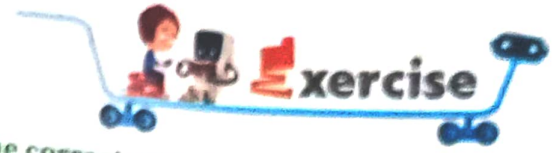
Step 3. Click the Print button to print your document.



Print Preview

Print Preview lets you see how your document will look when printed. To use it, click the Print Preview button on the Quick Access Toolbar.





I. Tick (✓) the correct option :

1. Which of the following buttons is used to align the text to the left?
   (a) ☐ (b) ☐ (c) ☐ (d) ☐
2. Text Effects are available in the ____ group on the Home tab.
   (a) Clipboard (b) Paragraph
   (c) Font (d) None of these
3. The ☰ option is used for creating a ____ list.
   (a) unordered (b) bulleted
   (c) numbered (d) none of these
4. Which of the following options allows you to apply some pretty cool effects to your text?
   (a) Print (b) Text effects
   (c) Home (d) Change case

II. Fill in the blanks with the help of the given box :

1. ..... means darker text.
2. In ..... alignment, the text is aligned along the right margin.
3. The ..... key is pressed twice for finishing the bulleted or numbered list.
4. The process of changing the appearance and arrangement of the text is called .....
5. A ..... is a small symbol such as square or a dot, used to mark each item in a list.

bullet
formatting
bold
enter
right

III. Match the following :

1. Ctrl + B	(a) Open
2. Ctrl + I	(b) Print
3. Ctrl + Z	(c) New Document
4. Ctrl + O	(d) Italic
5. Ctrl + C	(e) Undo
6. Ctrl + V	(f) Bold
7. Ctrl + N	(g) Copy
8. Ctrl + P	(h) Paste

IV. Answer the following questions :

1. What is font? Name any two fonts.
2. What is a bullet?
3. Differentiate between the following terms :
   (a) Bold and Underline
   (b) Editing and Formating
4. What do you mean by line spacing? Write steps to change the line spacing.
5. Write steps to Highlight the text.

Smart Thinking

Identify the each task of Word 2016 and write the correct name.

1. To perform a particular task. .....
2. To apply some pretty cool effects. .....
3. To apply different colours. .....
4. To improve the appearance of text. .....

Lab Activity

Open a new Word document and type as shown below :

Font : Times New Roman
Font Size : 12

Font : Broadway
Font Size : 18

Virat Kohli

Virat Kohli is an Indian cricketer who currently captains the India national team. He is a right handed top order batsman popular as one of the best batsmen in the world. He plays for Royal Challengers Bangalore in the Indian Premier League, and has been the team's captain since 2013.

He was born on 5 November 1988. Virat is the only current player to have a stand named after him in Delhi Firoz Shah Kotla stadium to honor his immaculate contributions to Indian cricket.

Font : Cambria
Font Size : 12

- Highlight underlined text.
- Add two lines about Virat Kohli and make these lines bullet points.

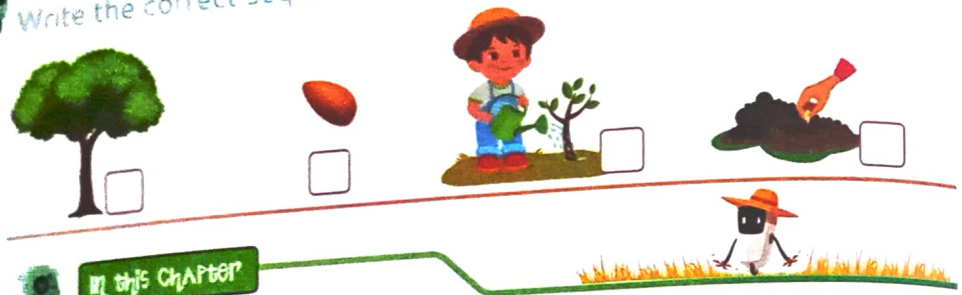
FOR THE TEACHER

- Discuss the advantages of using the Bullets and Numbering feature.
- Discuss the difference between editing and formatting.

7 Introduction to Coding

Let's RECALL

Write the correct sequence for growing a tree.



In this CHAPTER

- What is Coding?
- Designing a ScratchJr Project
- ScratchJr
- Components of the ScratchJr Window
- Projects

WHAT IS CODING?

Last year we learned that a program is a set of instructions that a computer follows to perform a task.

Coding is writing a set of these instructions that tell a computer or a machine what to do. Just like you give instructions to your friend to play a game, coding gives instructions to a computer.

Why is Coding Needed?

Coding is needed because it helps us create and control technology. We use coding to:

Build Programs and Apps : Make games, apps and websites.

Automate Tasks : Make machines and computers do tasks for us automatically.

Solve Problems : Find solutions to problems in an organised way.

Create New Things : Invent new gadgets, software, and systems that make our lives easier and more fun.

SCRATCH JR

ScratchJr is a programming language designed for young children, to create their own interactive stories and games.

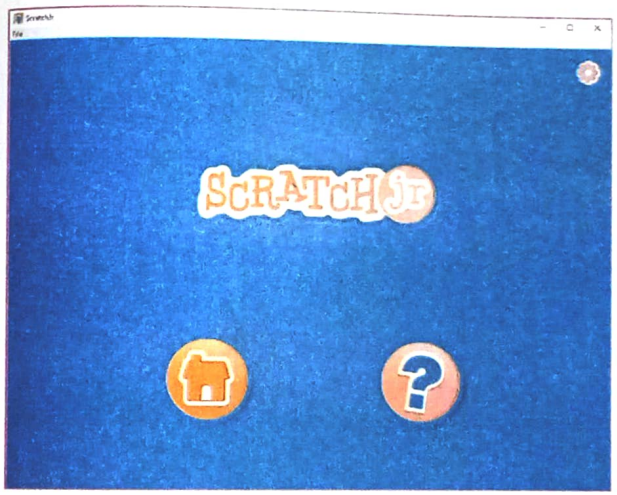
Instead of writing code, ScratchJr uses colorful puzzle-piece shapes. Kids drag and drop these shapes into the programming area and connect them to make characters move, jump, and dance. This helps them learn coding concepts while being creative and using their logical thinking.

ScratchJr is free to use and can be downloaded from the internet to install on your computer.

To open ScratchJr, follow these steps :

Step 1 : Click on the ScratchJr icon if it's on your desktop or taskbar. This will open the program.

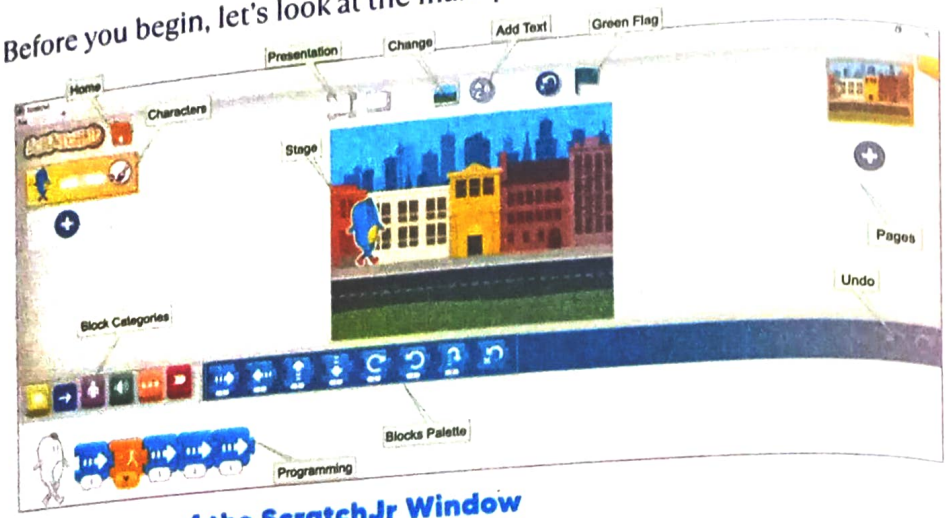
Step 2 : Click the Home button to see the main ScratchJr window. Now, you're ready to start your project.



INFO Byte

ScratchJr was developed at MIT college Boston with support from the National Science Foundation.

Before you begin, let's look at the main parts of the ScratchJr window :

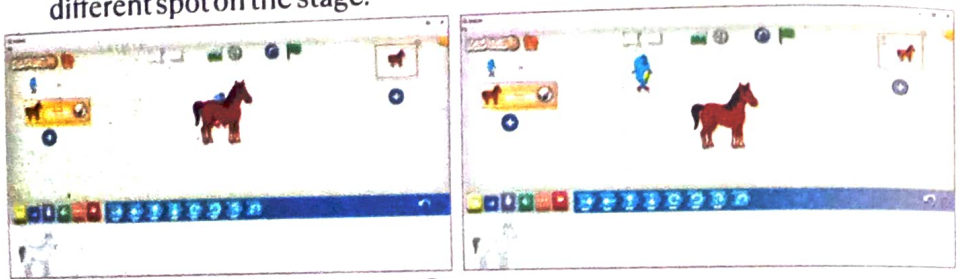


Components of the ScratchJr Window

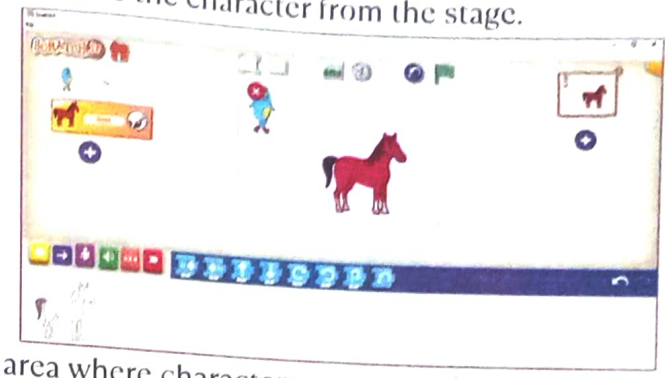
Add Characters

- Click the '+' button to see a list of different characters. Choose one by clicking on it, and it will appear in the middle of the stage.

- If a new character overlaps with an existing one, you can drag it to a different spot on the stage.



- To delete a character, press and hold it until an '×' sign appears. Click the '×' to remove the character from the stage.



Stage : The area where characters appear and perform the actions, when the code is run or played.

Character : The object on the stage that performs actions. You can choose different characters from the Add Characters option. The character that you see on the stage is referred to as the default character. The name of this character is Tic.

Programming Area : The area located just below the coding blocks is called programming area. You drag and connect blocks to create your code in this area.

Green Flag : It is the first block you connect in ScratchJr to start your code.

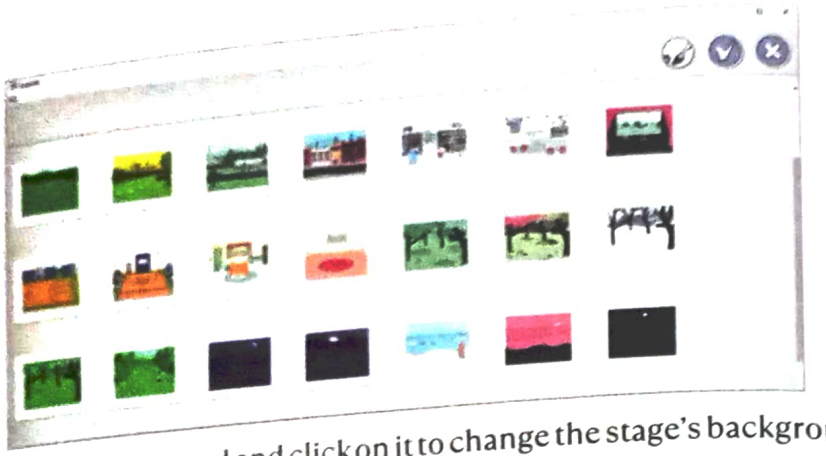
Presentation Mode : This mode is used to run and view your project. Click on the Presentation Mode button and then the Green Flag to start.

Add Background

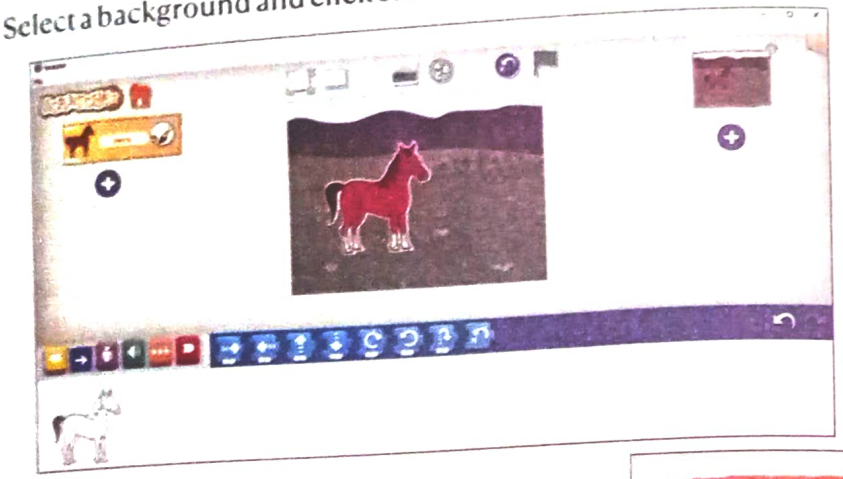
- The stage starts with a white background. Click the Add Background button above the stage to see a list of backgrounds.

Let's Get BETTER

If you could make a character do anything by giving it instructions, what would you make it do, and which steps would you use to make it happen?

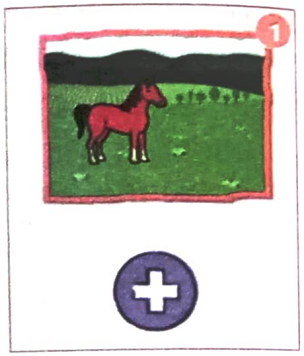


- Select a background and click on it to change the stage's background.

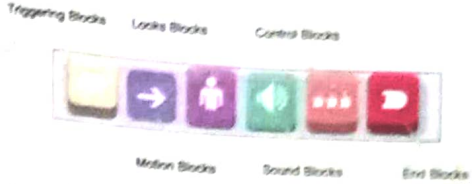


Add Projects

- In ScratchJr, a project is a set of coding blocks connected to perform actions.
- To add a new project, click the '+' button on the right side of the stage.

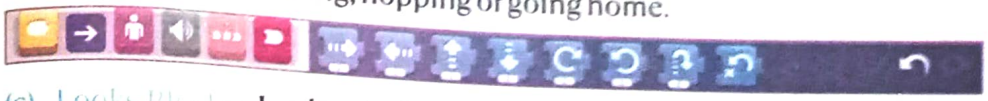


Block Categories : Blocks are used to give instructions to characters. They are grouped into categories that you can see on the right side of the screen when you click on the Block Category button. Different block categories are as follows:

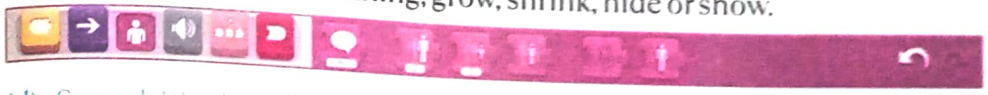


(a) Triggering Blocks : Triggering blocks start scripts and send messages between scripts.

(b) Motion Blocks : It control character movements such as moving right, left, up, down, turning, hopping or going home.

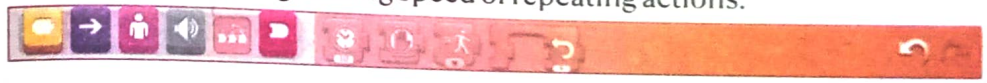


(c) Looks Blocks : Looks Blocks change how a character looks or appears, allowing it to say something, grow, shrink, hide or show.



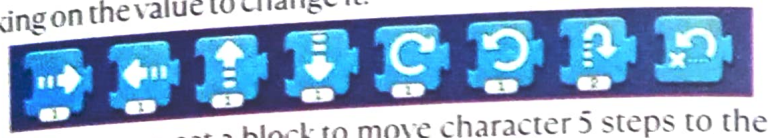
(d) Sound Blocks : Sound blocks play sounds like a pop or any sound recorded by the user.

(e) Control Blocks : It manages character actions based on conditions, like waiting, stopping, setting speed or repeating actions.

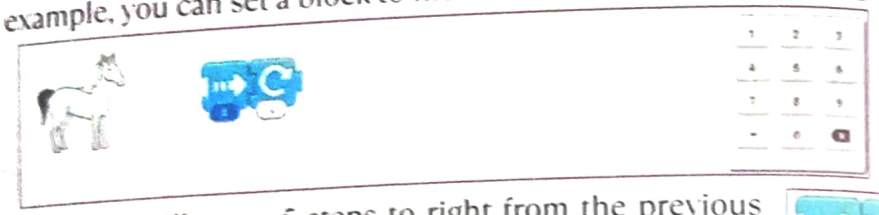


(f) End Blocks : It marks the end of a script to show when it's finished.

Coding Blocks : These blocks give commands to the characters. You can adjust the values in the blocks by dragging them to the programming area and clicking on the value to change it.



For example, you can set a block to move character 5 steps to the right.



The character will move 5 steps to right from the previous position.

Designing a ScratchJr Project

General Instructions :

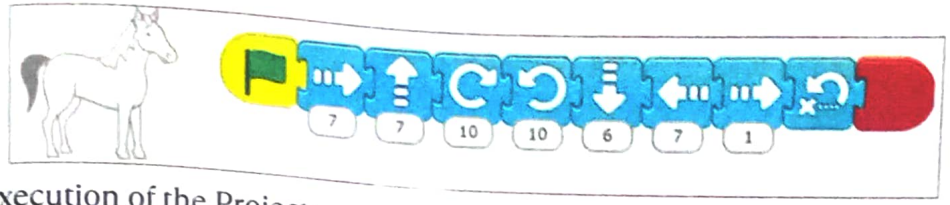
1. Start your project with the Green Flag.
2. Choose and place your character on the stage.
3. Select coding blocks from the categories.
4. Drag the blocks into the Programming Area.
5. Connect the blocks according to your plan.
6. Use the End Block to finish the code.
7. If you have more than one character, repeat the steps for each one.
8. Click on Presentation Mode.
9. Click the Green Flag to run your code.

Your project is now ready to go!

Project 1 : Moving and Rotating the Object

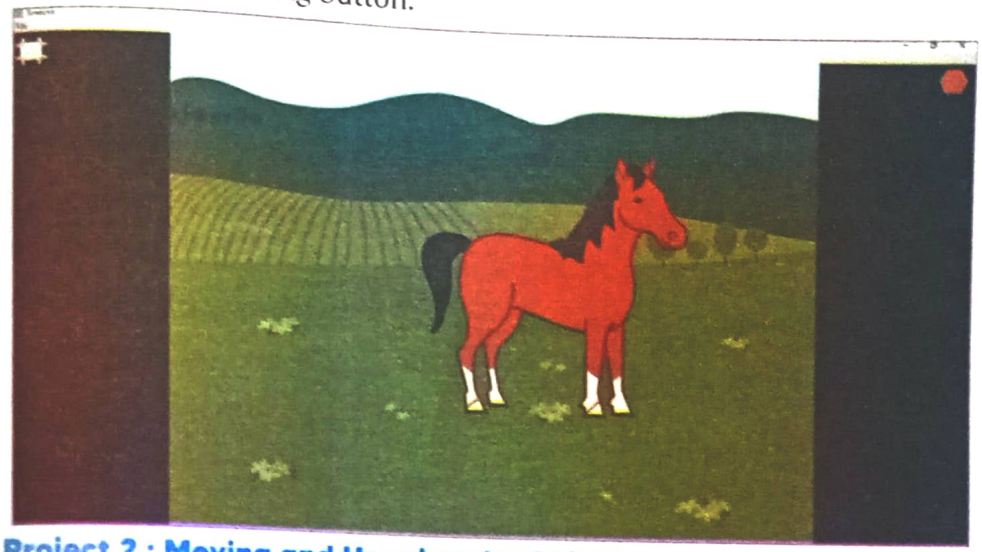
Make a character move and rotate on the stage using one character. Afterward, the character should return to its starting position.

Coding Blocks for the C[illegible]er :



Execution of the Project :

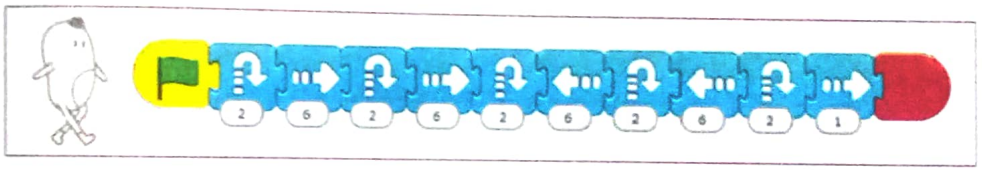
- Click on presentation mode button.
- Click on green flag button.



Project 2 : Moving and Hopping the Object

Make a character move and hop on the stage using only one character. Afterward, the character should return to its starting position.

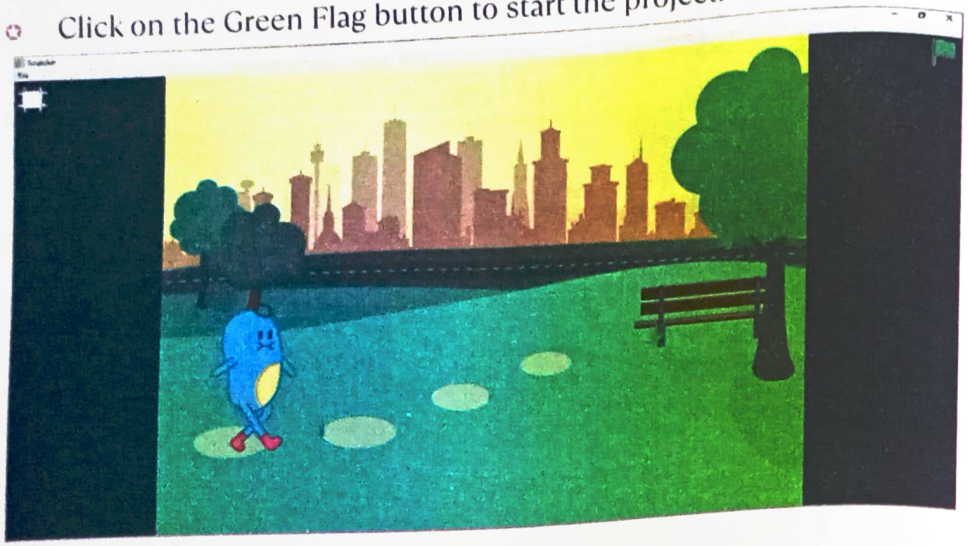
Coding Blocks for the Character :



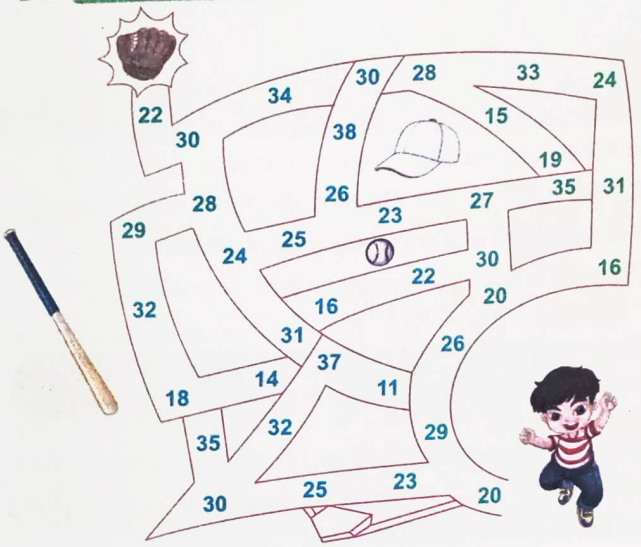
Execution of the Project :

- Click on the Presentation Mode button.

- Click on the Green Flag button to start the project.



Activity Corner

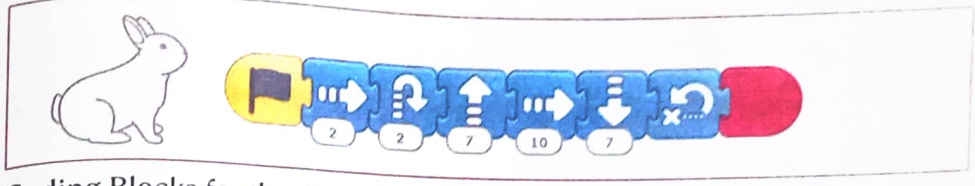


Lead Akash to his baseball glove by following any number between 20 and 30!

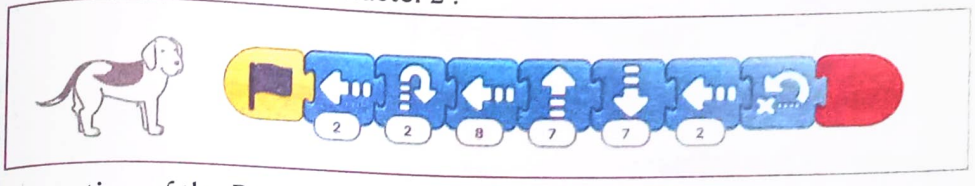
Project 3 : Crossing Two Objects

Use two characters to move across the screen from left to right and then return to their starting positions.

Coding Blocks for the Character 1 :



Coding Blocks for the Character 2 :



Execution of the Project :

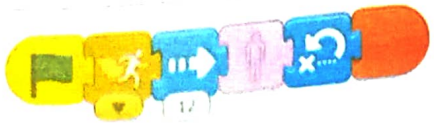
- Click on the Presentation Mode button.
- Click on the Green Flag button to start the project.

Project 4 : Cycle Racing

Use two characters to race across the screen from left to right.

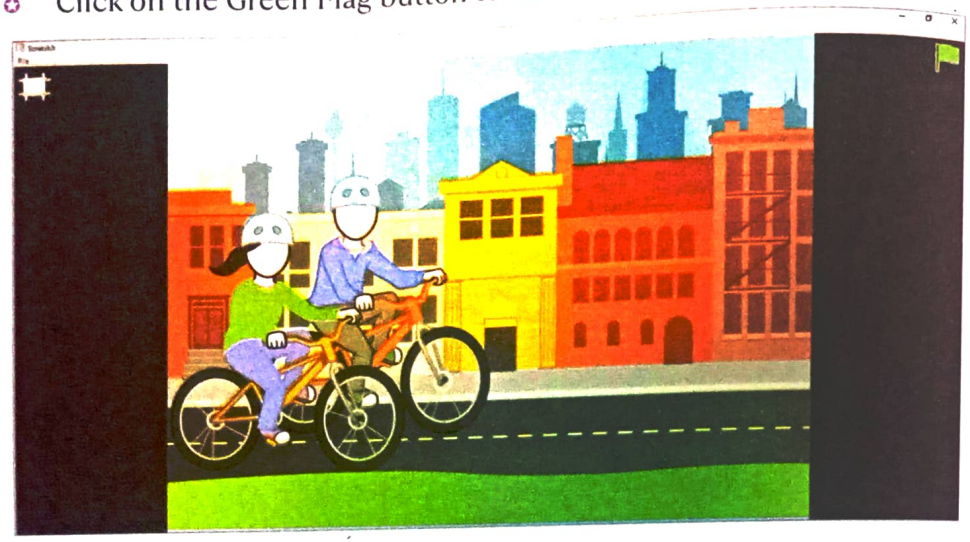
Coding Blocks for the Character 1 :

Coding Blocks for the Character 2 :



Execution of the Project :

- Click on the Presentation Mode button.
- Click on the Green Flag button to start the project.



I. Tick (✓) the correct option :

1. In ScratchJr, what type of action will be performed when the given block is clicked?

   (a) The object will move 5 steps to right. ☐
   (b) The object will move 5 steps to left. ☐
   (c) The object will go to home. ☐
   (d) The object will jump to 5 steps. ☐

2. Which of the following buttons is clicked to open ScratchJr window?
   (a) Go button ☐ (b) Start button ☐
   (c) Home button ☐ (d) House button ☐

3. Where do you drag and drop the coding blocks in ScratchJr programming?
   (a) Programming area ☐ (b) Stage area ☐
   (c) Execution area ☐ (d) Block area ☐

4. ScratchJr is a ..... visual programming.
   (a) based-block ☐ (b) shape-based ☐
   (c) block-based ☐ (d) design-based ☐

5. ..... flag is used as the first block in ScratchJr programming.
   (a) Green ☐ (b) Red ☐
   (c) Blue ☐ (d) Yellow ☐

6. ..... is a platform where the characters appear when the code is run.

(a) Project ☐ (b) Stage

(c) Structure ☐ (d) Construction

II. Write 'T' for true and 'F' for false :

1. In ScratchJr, we can create our own interactive stories and games. ☐
2. Coding blocks are the puzzle-piece shapes. ☐
3. It is very difficult to add background to make the stage more presentable. ☐
4. In ScratchJr, the coding blocks are connected vertically in the programming area. ☐
5. Presentation mode is used to run the project on full screen. ☐

III. Match the following :

Block Category	Name of the Block
	Motion
	Looks
	Sound
	Control
	Triggering

IV. Mention the type of a actions the following coding blocks will perform :

: .....

: .....

: .....

: .....

: .....

Smart Thinking

Example : V/52 E/15 R/43 Y/55 N/34 I/24 C/13 E/15

Learning is like a

☐/45 ☐/43 ☐/15 ☐/11 ☐/44 ☐/51 ☐/43 ☐/55

that will follow you

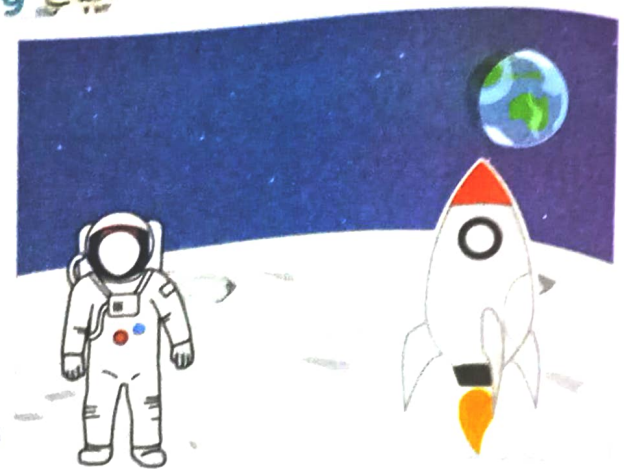
☐/15 ☐/52 ☐/15 ☐/43 ☐/55 ☐/53 ☐/23 ☐/15 ☐/43 ☐/15

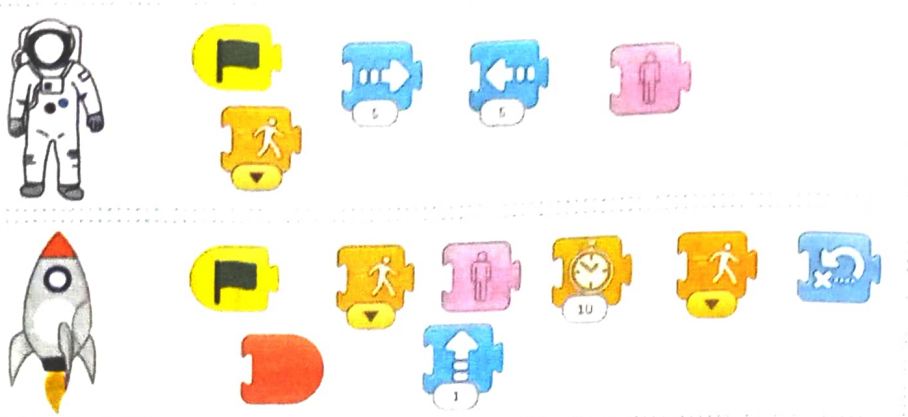
	1	2	3	4	5
1	A	B	C	D	E
2	F	G	H	I	J
3	K	L	M	N	O
4	P	Q	R	S	T
5	U	V	W	X	Y
6	Z				

Lab Activity

Choose two objects such as an astronaut and a rocket. When the astronaut approaches the rocket, it takes off.

Here, the coding blocks are defined in the programming area. Connect the blocks for both the objects and run the code.





Enjoy the thrill when executed!

FOR THE TEACHER

- Help students to open ScratchJr application.
- Show students how to drag and drop the Move and Loop blocks. Let them try different settings and see how these blocks change the character's movement.

CHAPTER 8

Artificially Intelligent Gadgets

LET'S RECALL

Write the names of voice assistants shown below.

IN THIS CHAPTER

- AI in Gadgets/Devices
- Artificial Intelligence in Homes
- Sensor
- Actuator
- Artificial Intelligence in Daily Life

In the previous class, you probably felt the excitement of being surrounded by artificial intelligence. It's amazing how advanced technology has transformed our lives. You might have noticed how AI learns, thinks, and acts like humans. In this lesson, we'll dive deeper into these smart devices and gadgets that help us manage our surroundings and work. Let's start by exploring these intelligent gadgets.

AI IN GADGETS/DEVICES

We use many AI-powered gadgets in our daily lives. For example, smart robot vacuum cleaners and smart thermostats are common AI devices that

make life easier and more comfortable. Have you ever wondered how these gadgets respond to their surroundings?

These devices use sensors to do their jobs. Two key terms you should know are sensors and actuators, which are important in artificial intelligence. We will learn about them before going into the main discussions.

Sensor

As you are aware, every machine works by processing the inputs that are fed to it in order to produce a desired output. Humans provide the inputs for machines in general. But what about machines with intelligence?

It is crucial to understand that a machine, whether intelligent or not, uses sensors to gather input before acting. Every single smart machine has the necessary sensors installed in order to gather data and information.

The responsive elements of an environment that receive input and cause the machine to operate are called sensors. These sensors could vary depending on the smart gadget. A few of them are listed below :

(i) Position Sensors
(ii) Pressure Sensors
(iii) Temperature Sensors
(iv) Vibration Sensors

Actuator

An actuator is a necessary component of a machine or device. Once the sensor provides input, it instructs the gadget to move, rotate, grab, and other physical activities. Stated differently, it is the part of any smart machine that makes movement possible.

For instance, different kinds of robots use actuators to carry out various tasks. Artificial intelligence is widely applied in many different contexts, including :

- residences
- daily life
- e-commerce
- social media platforms
- entertainment
- manufacturing industries

And so forth.

Now we'll focus on how AI is used in homes and in everyday life.

ARTIFICIAL INTELLIGENCE IN HOMES

There are many AI devices we use at home. Let's look at some examples :

AI Cleaners (Robot Vacuum Cleaners)

Cleaning floors? No problem! Robot vacuum cleaners can handle it for you. You just set instructions, like how many times to clean a room, and the robot takes care of the rest. Roomba is a popular choice for this.

Smart Assistants

A smart assistant helps you with tasks and answers questions. It can control other smart devices, like TVs and lights, using voice commands. Popular smart assistants include Alexa (Amazon), Siri (Apple), and Google Assistant.

Smart Cameras

Smart cameras are essential for monitoring your home. They can be placed inside or outside, connect to your Wi-Fi, and send data to your phone or cloud storage. They help you stay connected with loved ones, keep an eye on things, and provide 24/7 security.

Smart Thermostats

Feeling too hot or cold? A smart thermostat adjusts your home's temperature based on your settings. It saves energy by turning the heating or cooling on and off as needed and can be controlled remotely via your smartphone.

Let's Get Better

How do smart devices like robot vacuums and smart thermostats make life easier for people? Can you think of other ways these devices could help in the future?

Smart Door Locks

Smart door locks use fingerprints for access and can be controlled via Bluetooth or Wi-Fi. They're secure and convenient, and you can manage who has access.

ARTIFICIAL INTELLIGENCE IN DAILY LIFE

Robots

Robots are programmable machines that can perform tasks with little human help. They're used in various ways, such as :

Humanoid Robots : Like Pepper, which interacts with people and helps in stores.

Restaurant Robots : Like Penny, which transports and serves food in restaurants.

Business Robots : Like Promobot, which communicates, answers questions, and manages tasks in businesses.

Security Robots : Like Nimbo, which patrols areas, detects security issues, and alerts guards.

Self-Driving Cars

Once a dream, self-driving cars are now a reality thanks to companies like Tesla. These cars can drive themselves, sensing their surroundings and operating without human input. They make travel more convenient.

Smoke Detectors

Smoke detectors sense smoke and alert you to potential fires. They connect to Wi-Fi and can send alerts to your phone if you're not home. They're useful in homes, offices, and other buildings.

Google Lens

Google Lens

Google Lens uses your phone's camera to recognise objects and provide information. It can scan, translate, and search for items quickly. For example, pointing it at a dog or a product will give you details about them.

Artificial Intelligence : The Present and Future

AI has significantly changed our lives. From smartphones and internet use to online shopping and social media, AI impacts how we live. It's clear that AI is a big part of our future, shaping how we interact with technology.

I. Tick (✓) the correct option :

1. This popular smart assistant is known as .....
   (a) Siri ☐
   (b) Hey ☐
   (c) Alexa ☐
   (d) Bose ☐

2. ..... is not an AI enabled gadget.
   (a) iPhone ☐
   (b) Driverless car ☐
   (c) Robot ☐
   (d) Lens ☐

3. Which of the following statements is not true for a sensor in AI?
   (a) It takes an input
   (b) It reflects an action
   (c) It does not reflect an action
   (d) It is an AI enabled device

4. What is not appropriate for a smart car?
   (a) It does not require fuel
   (b) It is an intelligent car
   (c) It is a driver less car
   (d) It requires an assistant to drive a car

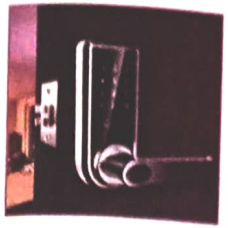
5. Choose the odd one out :
   (a) Camera ☐ (b) Google lens
   (c) Photographer ☐ (d) Light

6. Which of the following statements is appropriate for a smoke detector?
   (a) It is an entertainment device
   (b) It plays songs for you
   (c) It makes an alarm
   (d) It cleans walls of your home

II. Write 'T' for true and 'F' for false :

1. All AI enabled devices are man-made. ☐
2. Google lens is an AI enabled technology by Amazon. ☐
3. All smart devices are not intelligent. ☐
4. Roomba is one the most popular robot vacuum cleaners. ☐
5. The first industrial robot was invented by George Devol. ☐

III. Name the following AI enabled devices :



....

IV. Write short notes on each of the following :

1. Sensor (reference to AI)
2. Actuator (reference to AI)
3. Robot
4. Google Lens

V. Name the robot used for the following services :

1. Security Robot .....
2. Business Robot .....
3. Humanoid Robot .....

VI. Name three AI enabled devices which are used in homes.

VII. Name three AI-powered devices which are used in our daily lives.

Smart Thinking

1. It is assumed that you are living in a smart home. Now, enlist any four AI enabled devices that are available in your home.
   (a) .....
   (b) .....
   (c) .....
   (d) .....

2. You had gone to a weekend tour with your parents. You captured few photographs of the sight scene in your smart camera. Now, select any five pics. Using Google Lens, name the places that you have visited.

(a) .....

(b) .....

(c) .....

(d) .....

(e) .....

FOR THE TEACHER

- Allow students to explore AI gadgets through videos or demonstrations.
- Break down how sensors and actuators work with real life examples that are relatable to student.

Periodic Test Term-1

(Based on Chapters 1 to 3)

Time .....

Marks .....

Note : All questions are compulsory.

A. Tick (✓) the correct option :

1. A computer works on the ..... cycle.
   (a) PIO ☐ (b) IPO ☐ (c) IOP ☐ (d) POI ☐
2. ..... is usually fixed inside a digital camera.
   (a) Pen drive ☐ (b) Hard disk ☐
   (c) Memory card ☐ (d) None of these ☐
3. Which of the following is not a storage device?
   (a) CD ☐ (b) Pen drive ☐
   (c) Hard disk ☐ (d) Laser printer ☐

B. Write 'T' for true and 'F' for false.

1. Input is the result we get after processing. ☐
2. The output which is displayed on the monitor is called hard copy. ☐
3. Removable disk means removing hard disk from computer. ☐

C. Circle the odd one out :

1.	Paint	Word	Windows 10	Adobe Photoshop
2.	Mouse	Scanner	Keyboard	Windows XP
3.	CD	Pen drive	Hard disk	Keyboard

D. Answer the following questions :

1. What do you mean by output? Write the names of any two output devices.
2. What is an operating system? Name any one operating system used in your school's computer lab.

Half Yearly Test Paper

(Based on Chapters 1 to 5)

Time .....

Marks .....

Note : *All questions are compulsory.*

A. Tick (✓) the correct option :

1. Which of the following is an input device?
   (a) Keyboard ☐ (b) Printer ☐
   (c) CPU ☐ (d) Speaker ☐
2. Which of the following tools is used to see the enlarged view of the drawing area?
   (a) Magnifier ☐ (b) Magnifying Glass ☐
   (c) Power Glass ☐ (d) Lens ☐
3. In Paint, the file extension .bmp stands for ..... .
   (a) bit map project ☐ (b) bitmap image ☐
   (c) base map project ☐ (d) basic map image ☐
4. Which of the following things cannot be done with the help of a computer?
   (a) Play games ☐ (b) Share feelings ☐
   (c) Solve sums ☐ (d) Shopping ☐
4. To select a sentence, click the left mouse button while holding the ..... key.
   (a) Ctrl ☐ (b) Shift ☐ (c) Alt ☐ (d) Enter ☐

B. Fill in the blanks :

1. ..... shows us the result after processing.
2. A computer accepts data and instruction from you through ..... .
3. To delete large text, ... it and press ..... key.
4. Making changes in the existing text is called ..... the document.

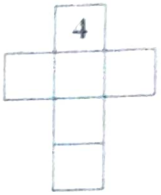
C. Write 'T' for true and 'F' for false.

1. The output which is displayed on the monitor is called hard copy. ☐
2. File is a storage device. ☐
3. Paint generally saves your drawing in .jpeg format. ☐
4. Status bar is located at the bottom of Word window. ☐

D. Write the numbers 2, 3, 5, 6 and 7 in correct place so that each line of the cross adds up to 15.

2 3 5 6 7

Total must be 15.



E. Ankit wants to throw a birthday party for his friends. He wants to invite them by writing an invitation letter. Which application software will he use to write the letter?

F. Write short notes on each of the following :

1. Copy and Paste
2. Free-form selection

G. Answer the following questions :

1. What are the Supercomputers?
2. What are processing devices?
3. Differentiate between copying and cutting text.
4. Write any two features of Word 2016.

H. Look at the given pictures and fill the missing letters :

Periodic Test Term-2

(Based on Chapters 6 and 7)

Time .....

Marks .....

Note : All questions are compulsory.

A. Tick (✓) the correct option :

1. Which of the following buttons is used to align the text to the left?
   (a) (b) (c) (d)
2. Text Effects are available in the ..... group on the Home tab.
   (a) Clipboard (b) Paragraph
   (c) Font (d) None of these
3. ScratchJr is a ..... visual programming.
   (a) based-block (b) shape-based
   (c) block-based (d) design-based

B. Write 'T' for True and 'F' for False :

1. In ScratchJr, we can create our own interactive stories and games.
2. Coding blocks are the puzzle-piece shapes.
3. It is very difficult to add background to make the stage more presentable.

C. Fill in the boxes :

1. ..... means darker text.
2. In ..... alignment, the text is aligned along the right margin.
3. The ..... key is pressed twice for finishing the bulleted or numbered list.

D. Differentiate between the following terms :

1. Bold and Underline
2. Editing and Formating

Annual Test Paper

Time .....

Marks .....

Note : All questions are compulsory.

A. Tick (✓) the correct option :

1. The option is used for creating a ..... list.
   (a) unordered (b) bulleted
   (c) numbered (d) none of these
2. Which of the following options allows you to apply some pretty cool effects to your text?
   (a) Print (b) Text effects
   (c) Home (d) Change case
3. Which of the following buttons is clicked to open ScratchJr window?
   (a) Go button (b) Start button
   (c) Home button (d) House button
4. What is not appropriate for a smart car?
   (a) It does not require fuel (b) It is an intelligent car
   (c) It is a driver less car (d) It requires an assistant to drive a car

B. Fill in the blanks :

1. The process of changing the appearance and arrangement of the text is called ..... .
2. A ..... is a small symbol such as square or a dot, used to mark each item in a list.
3. ..... flag is used as the first block in ScratchJr Programming.
4. ..... is a platform where the characters appear when the code is run.

C. Match the following :

1. Ctrl + B	(a) Open
2. Ctrl + I	(b) Print
3. Ctrl + Z	(c) New Document
4. Ctrl + O	(d) Italic
5. Ctrl + C	(e) Undo
6. Ctrl + V	(f) Bold
7. Ctrl + N	(g) Copy
8. Ctrl + P	(h) Paste

D. Write 'T' for True and 'F' for False :

1. Google lens is an AI enabled technology by Amazon.
2. All smart devices are not intelligent.
3. Roomba is one the most popular robot vacuum cleaners.
4. The first industrial robot was invented by George Devol.

E. Answer the following questions :

1. What do you mean by line spacing? Write steps to change the line spacing.
2. Write steps to Highlight the text.
3. Name three AI enabled devices which are used in homes.
4. Name three AI-powered devices which are used in our daily lives.

F. Identify the each task of Word 2016 and write the correct name :

1. To perform a particular task. .....
2. To apply some pretty cool effects. .....
3. To apply different colours. .....
4. To improve the appearance of text. .....

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AI AND ROBOTICS

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